

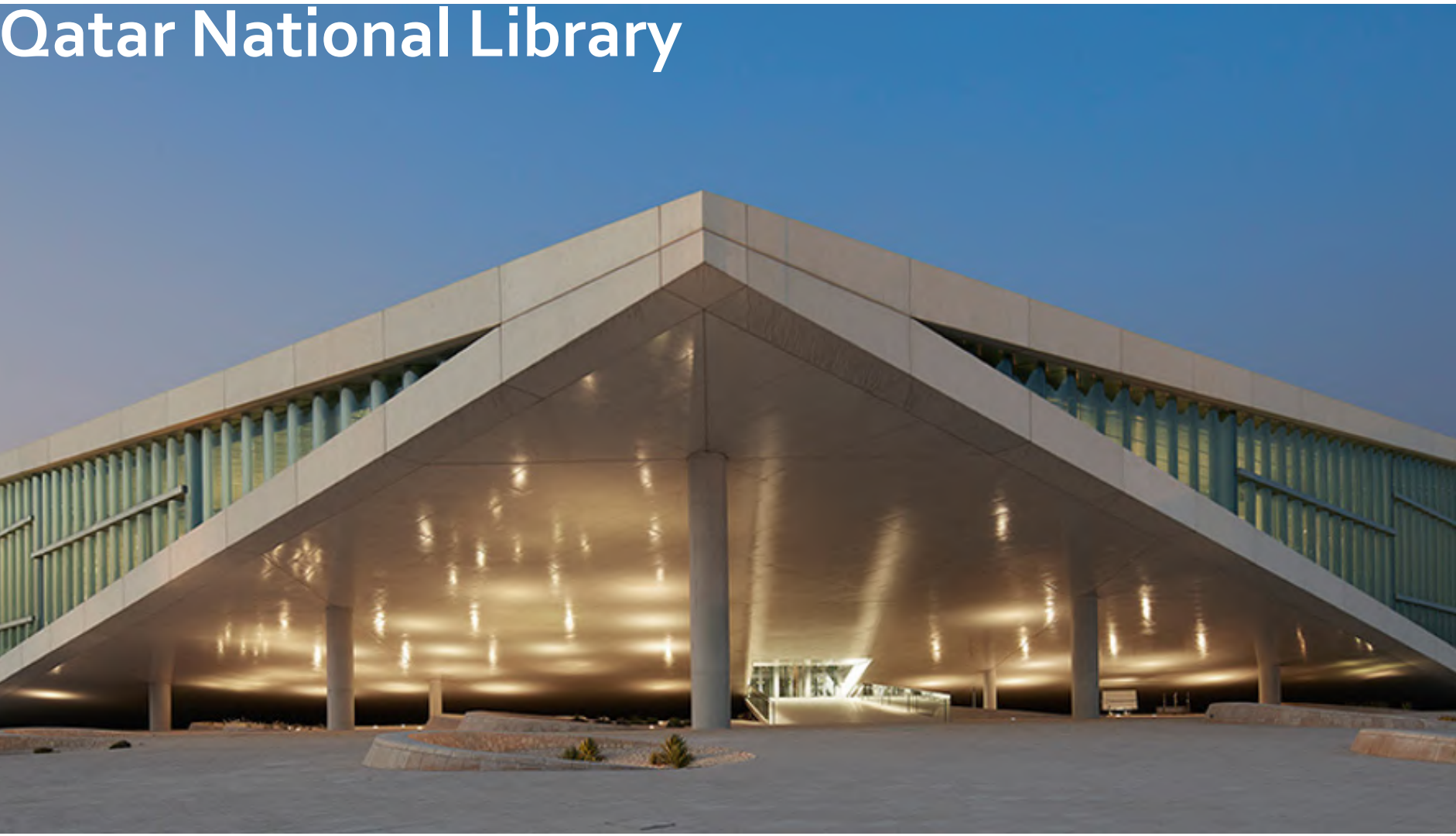


Architectural Design Portfolio

Design Project 02

Majd Bayakli - H00515431

Qatar National Library

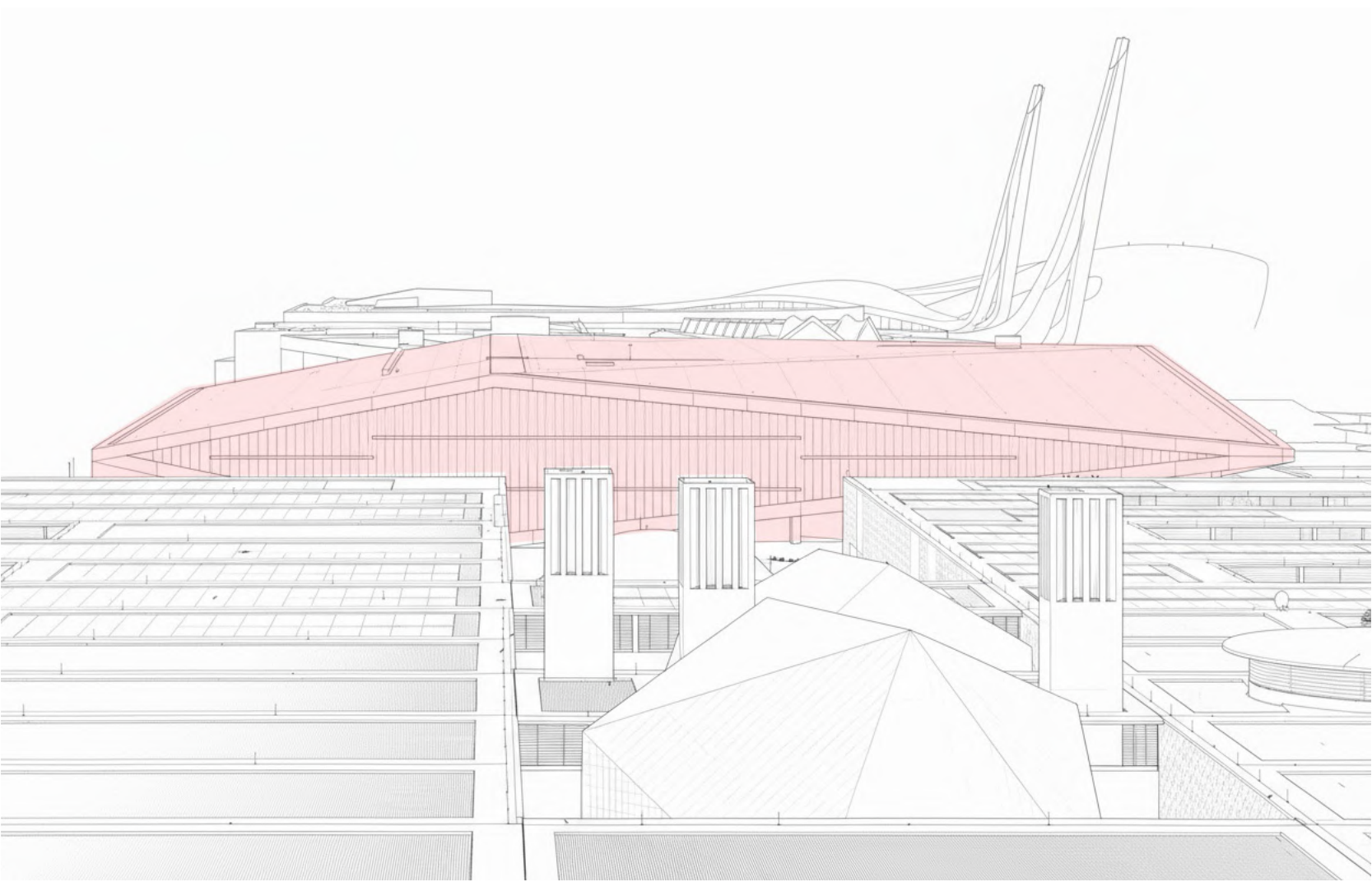


Doha, Qatar

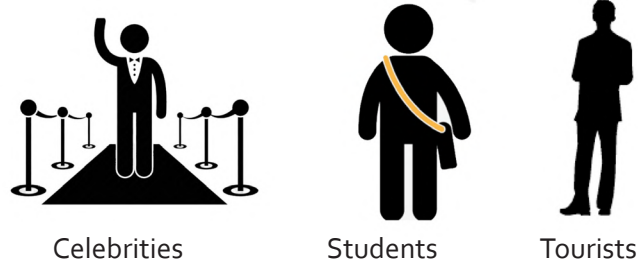
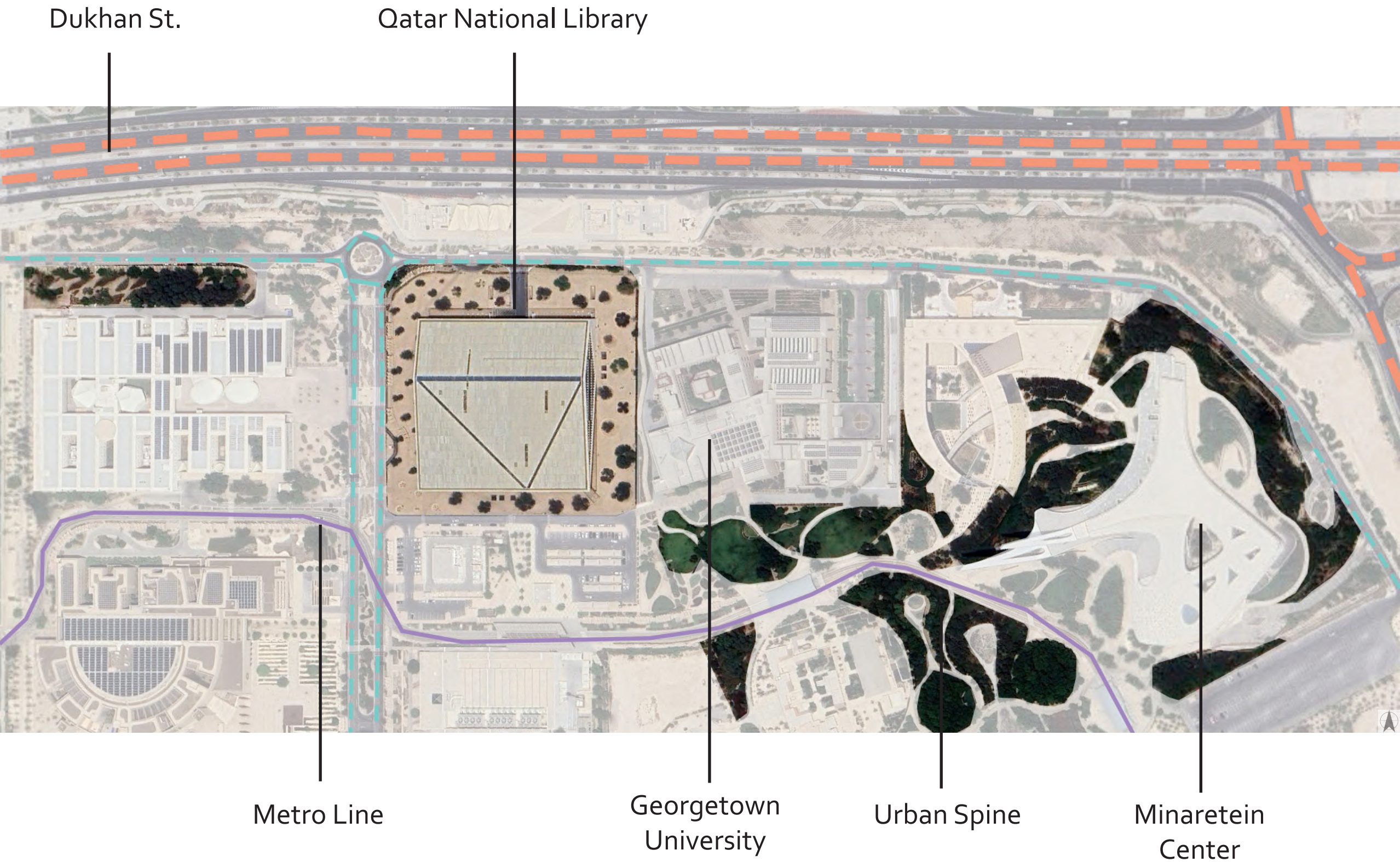
 Architects : OMA

 45000 sqm

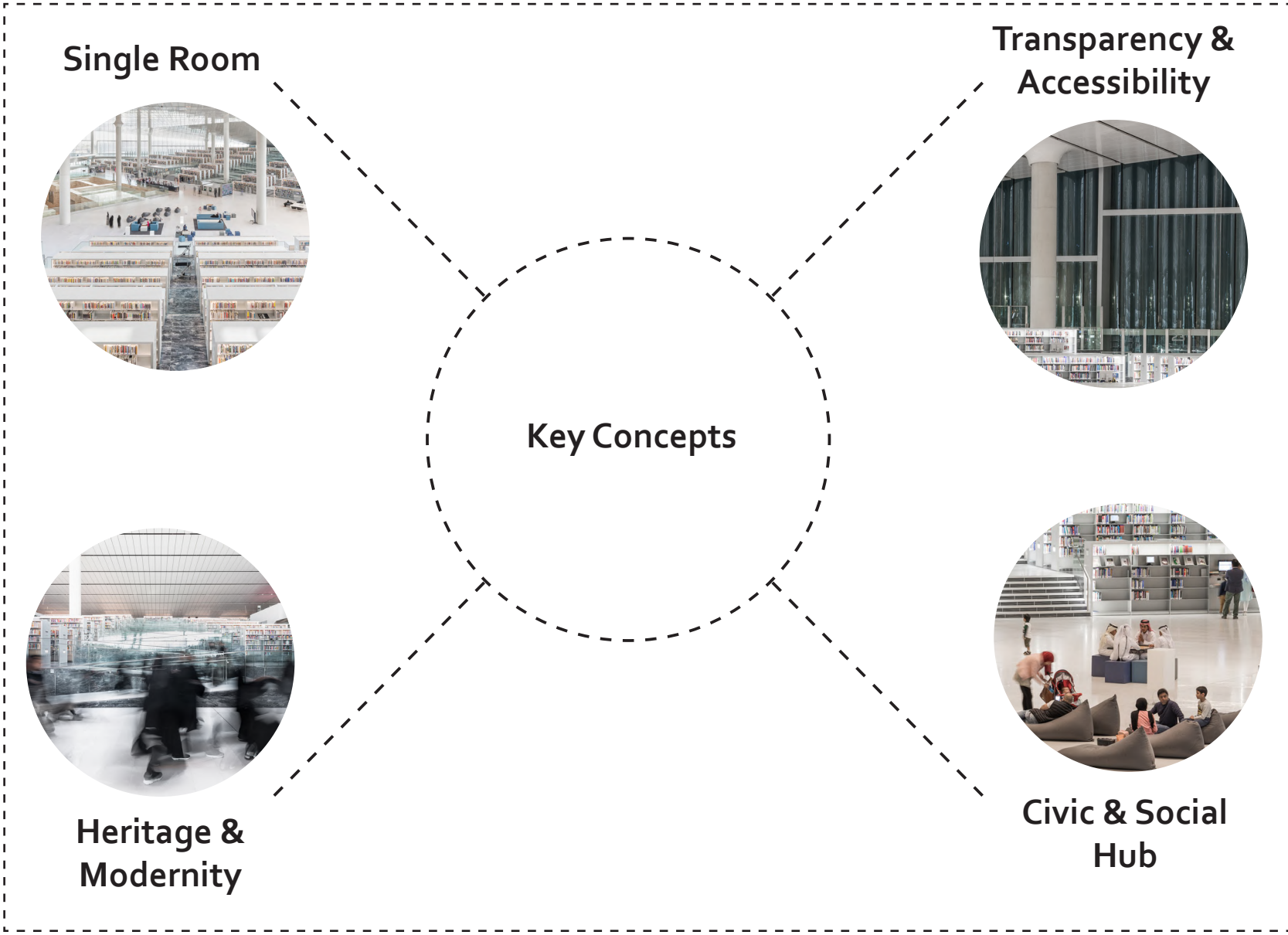
 2017



The Qatar National Library is one of the landmarks within education city in Doha, embodying Qatar's goal to use knowledge, research, and importantly culture exchange at the core of national development. Designed by Rem Koolhaas and executed by OMA, the library reideates the traditional typology of a library as an open and expansive design concept from within, dissolving boundaries between users, collections, and disciplines, encouraging interactions and intellectual experiences.



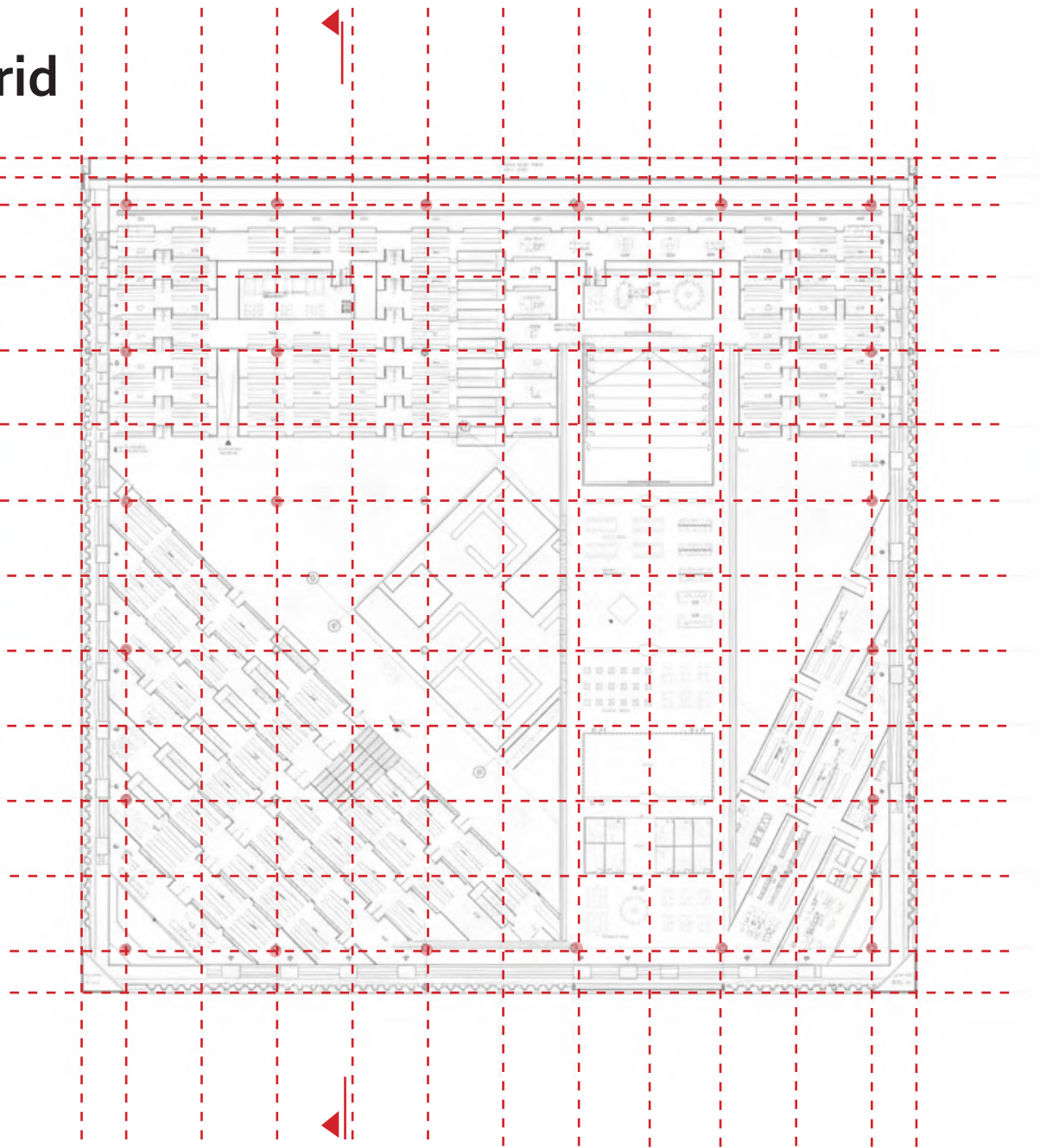
Many different types of users pass through the Qatar national library, while not only displaying the history of qatar, spaces are provided to create new and exciting projects.



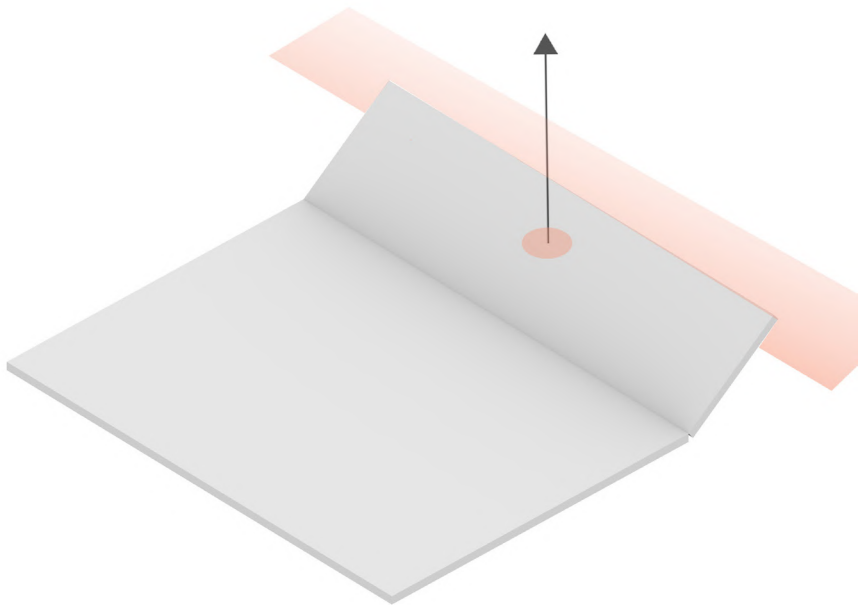
Construction



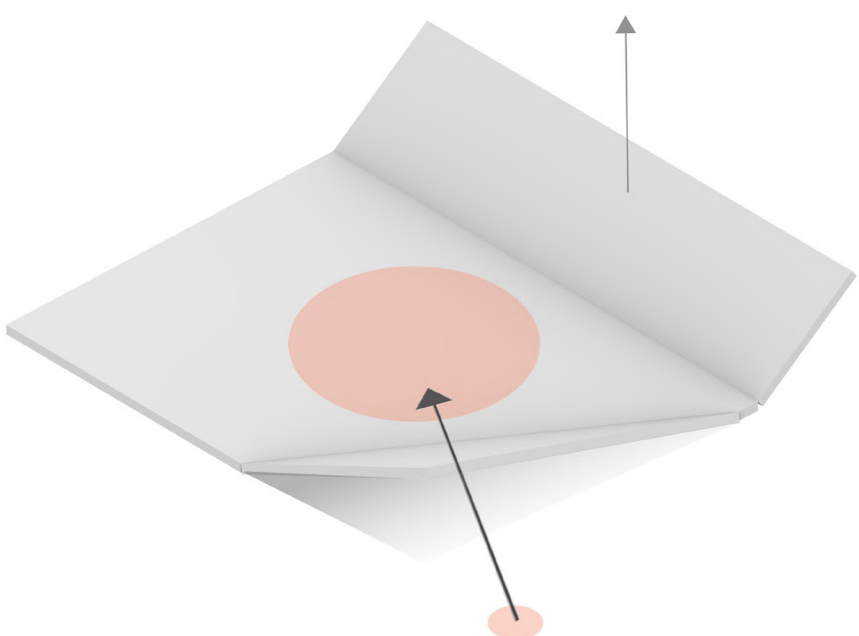
Structural Grid



Design Development



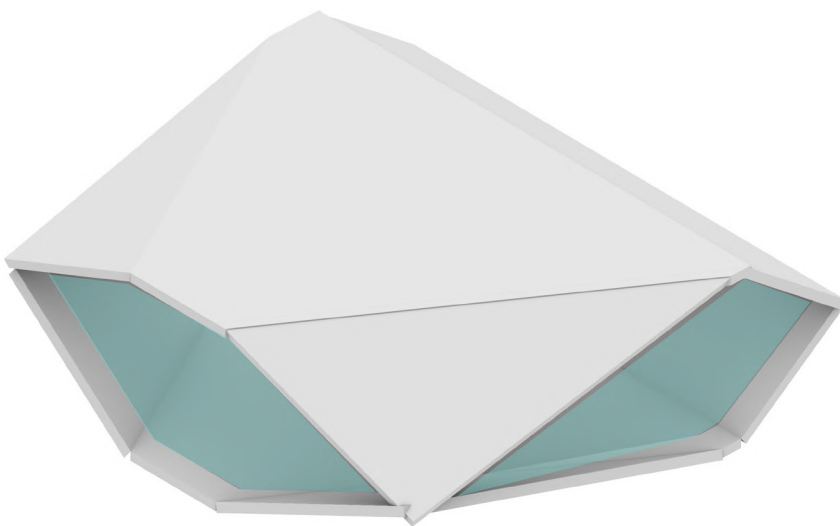
Elevated to block highway



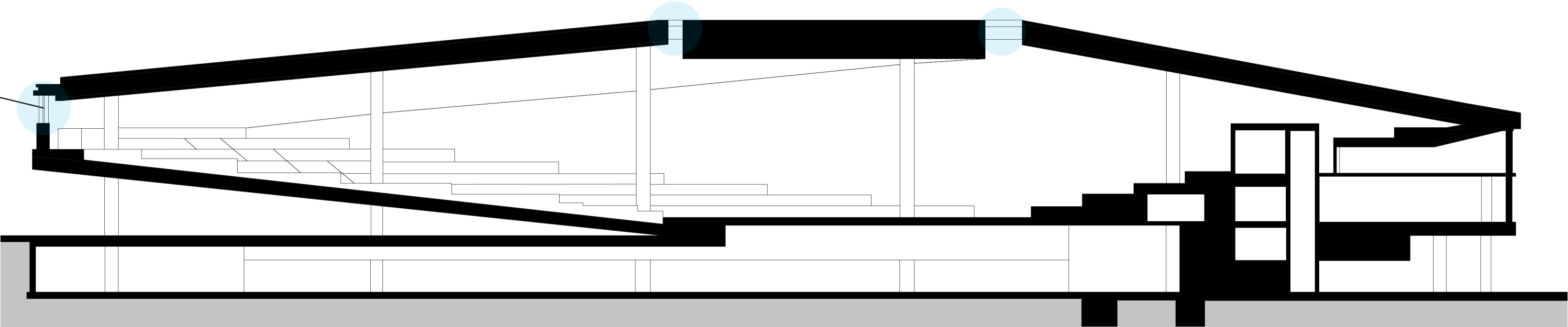
Entrance to main space



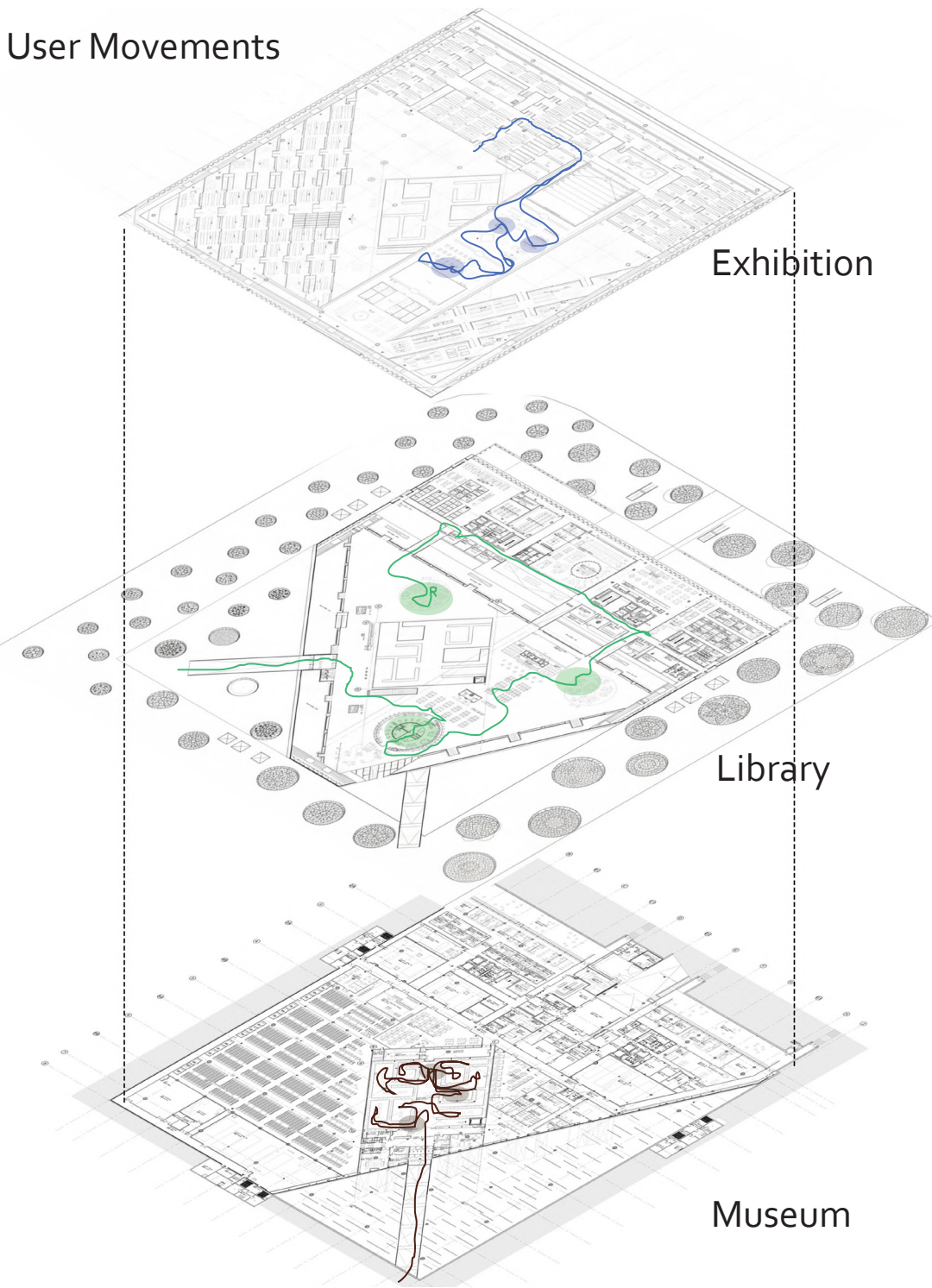
Mirrored roof structure



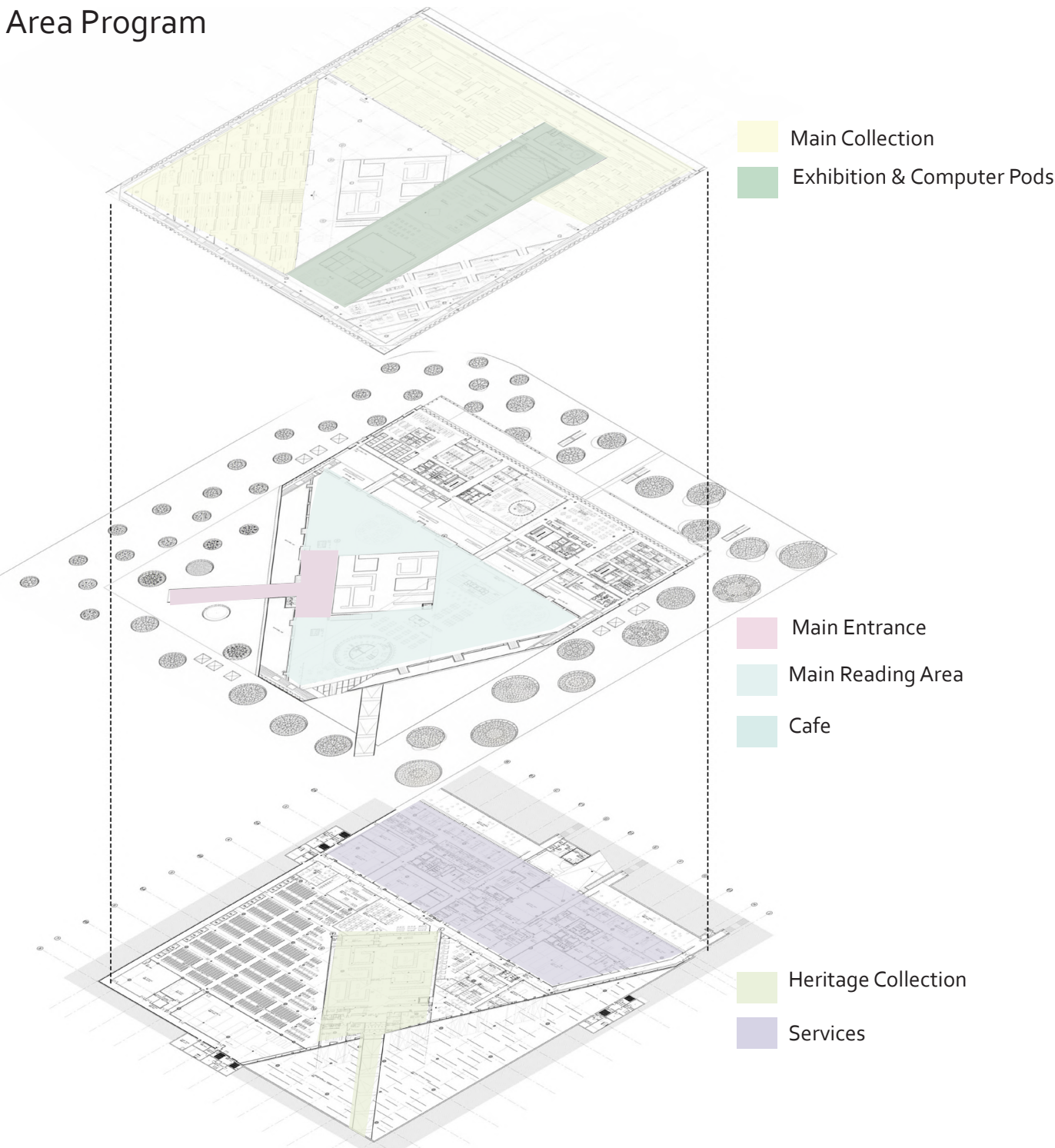
Diffused glare-free daylight facade



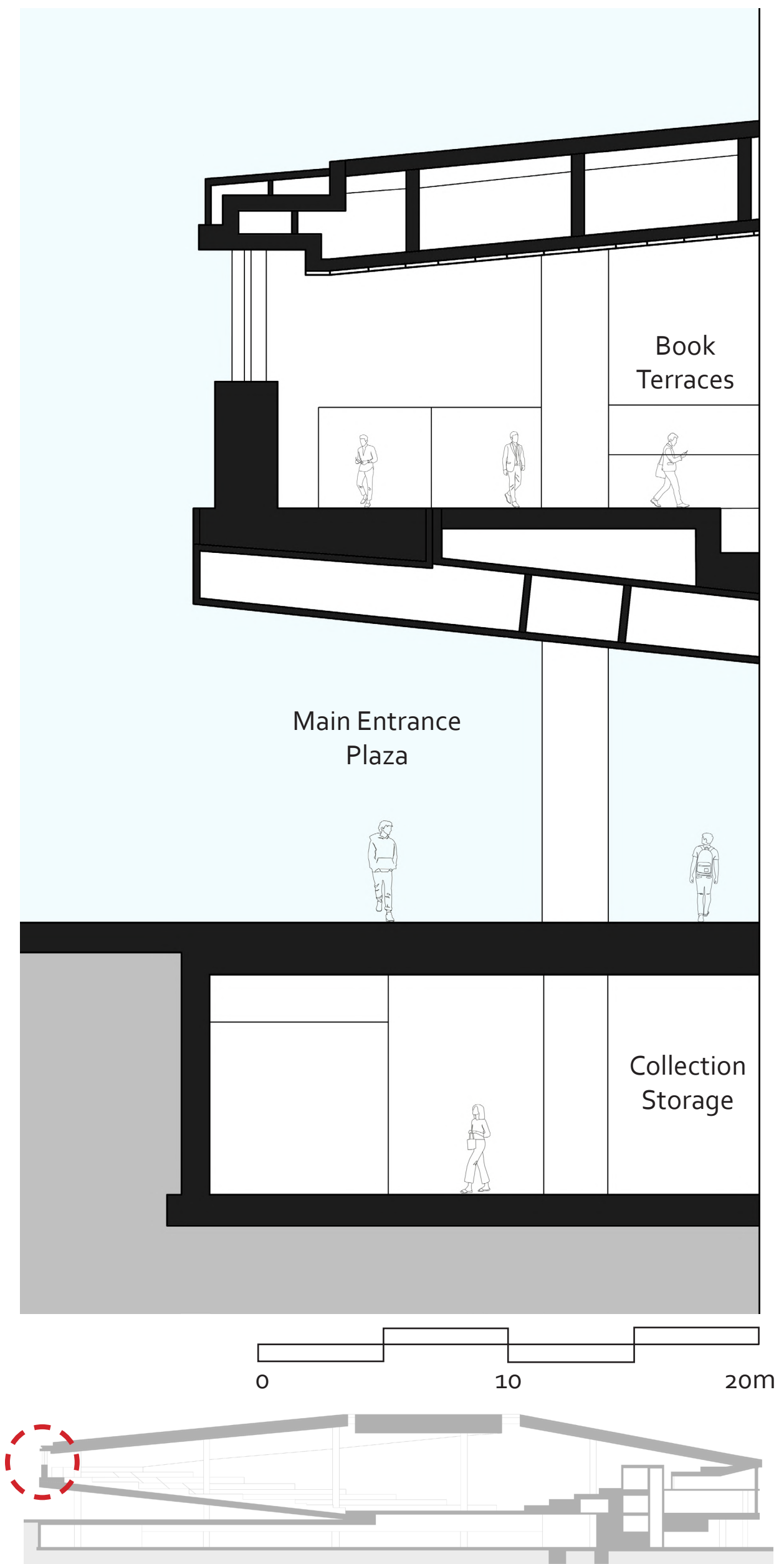
User Movements



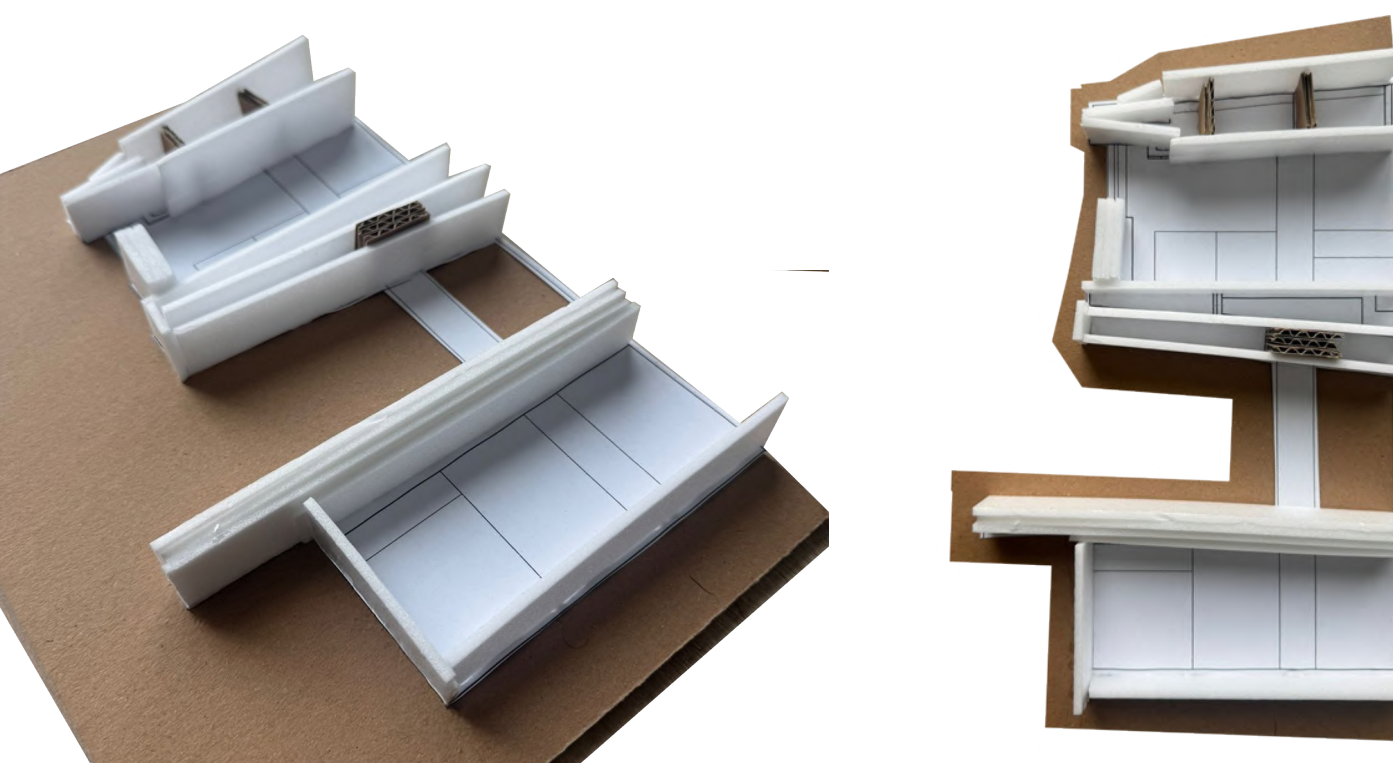
Area Program



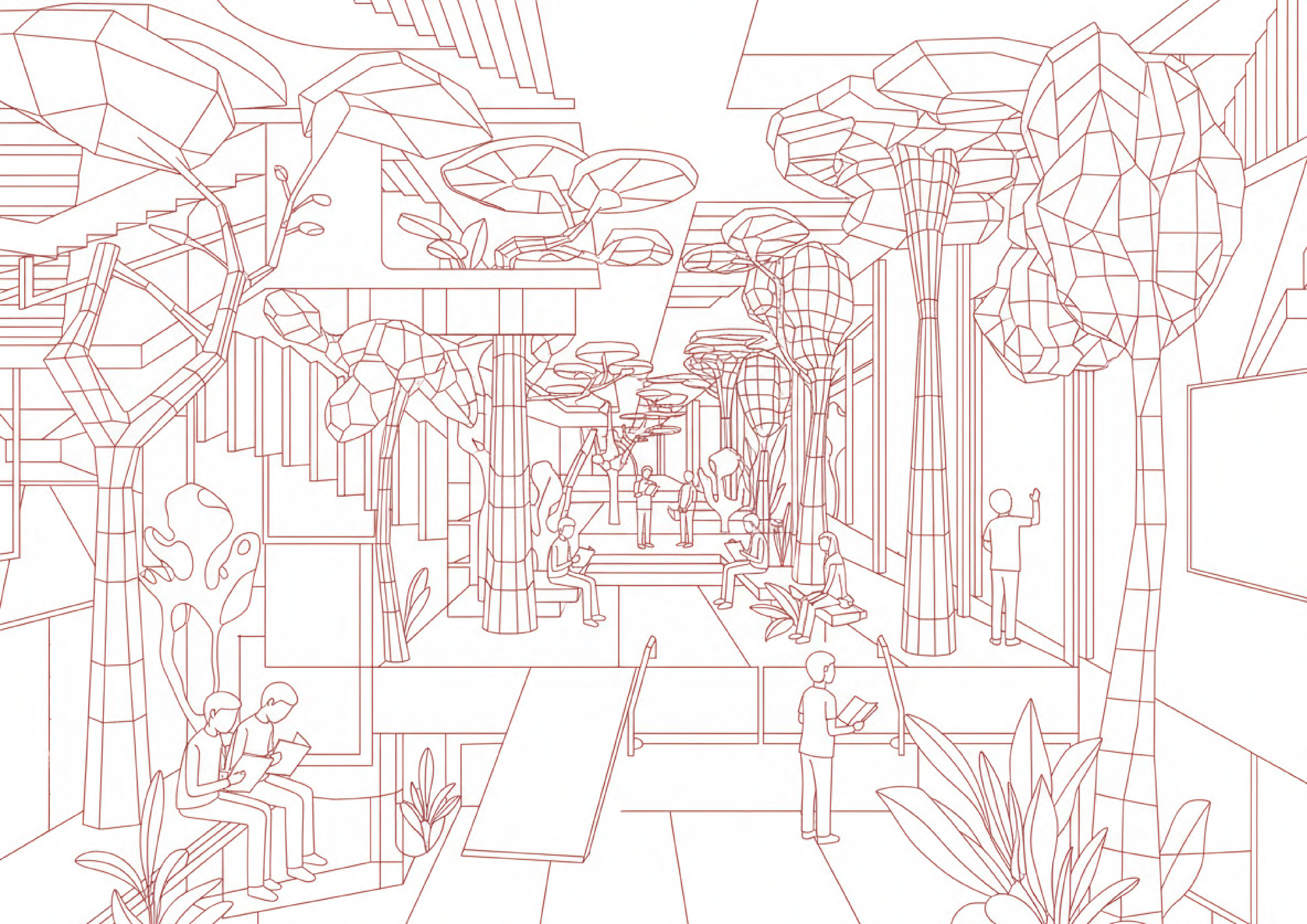
Section & Model Development



This model study of the Qatar National Library is expressed through a continuous geometric form that appears to surround a central space, making planes appear from the ground. This creates an interior experience where floor, wall, and platforms are read as one fluid architectural sense rather than separate elements. The form helps organize the library program, guides movement, and visually connects readers to seek for their knowledge. Using a single open spatial system.



The sectional model expresses the library as a floating edge architectural form, where the main volume is lifted from the ground to create a sense of lightness and separation. The structural system allows the floorplate to cantilever, reinforcing the geometry while making the building appear suspended rather than grounded. The visual identity of the design creates layered spatial relationships between low and upper levels, allowing movement, light, and openness through the section.





October 2003



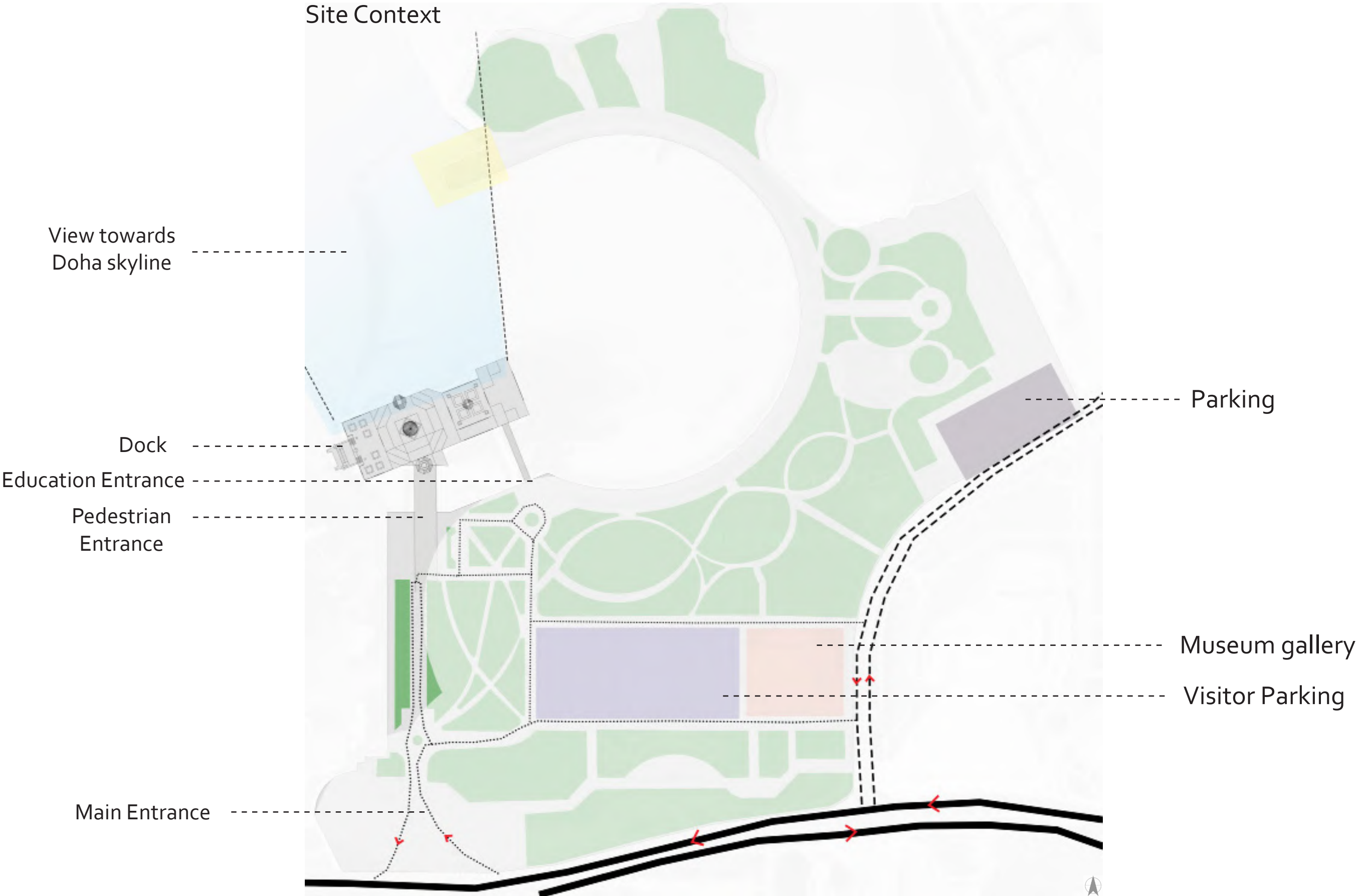
November 2005



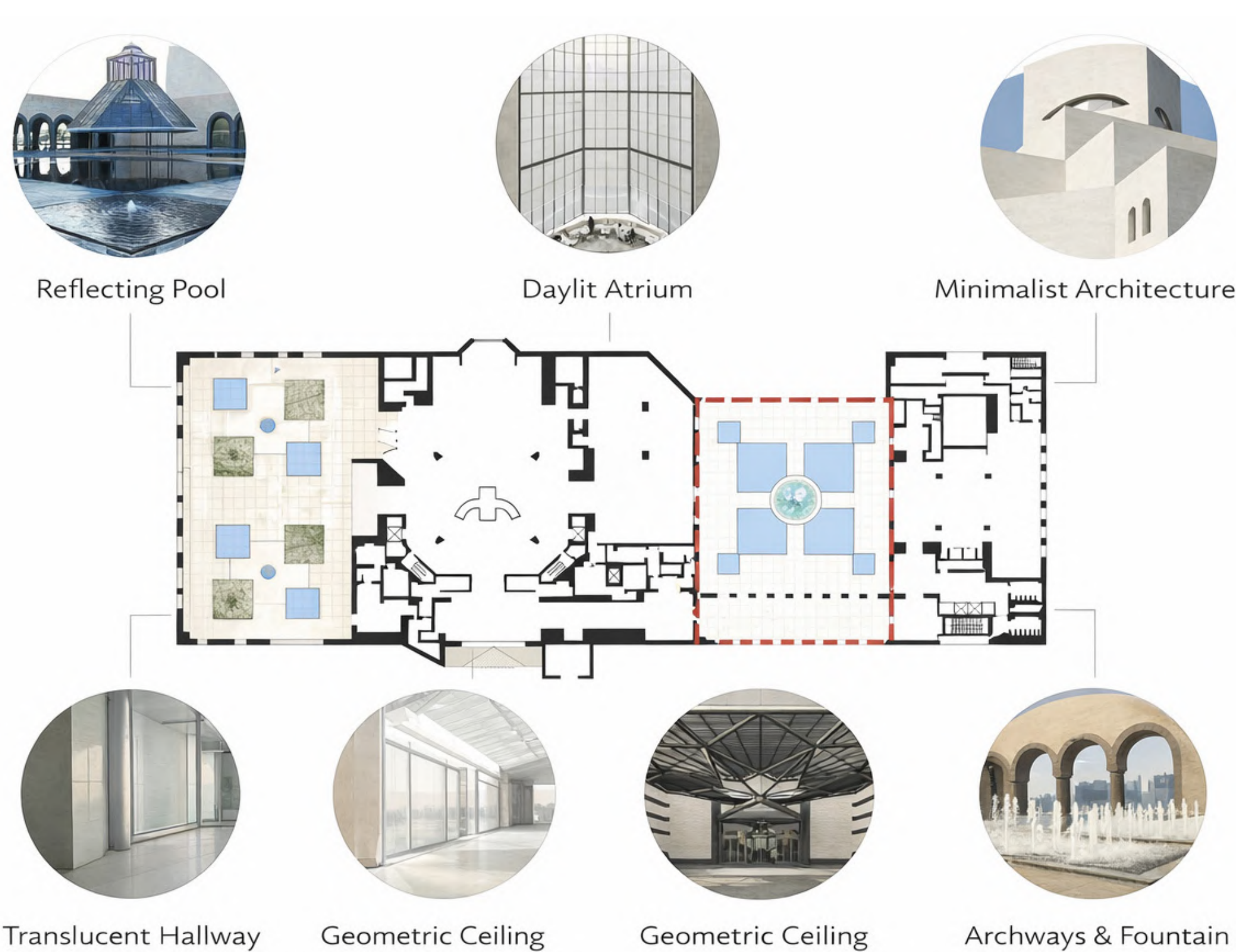
July 2009

The Museum of Islamic Art in Doha

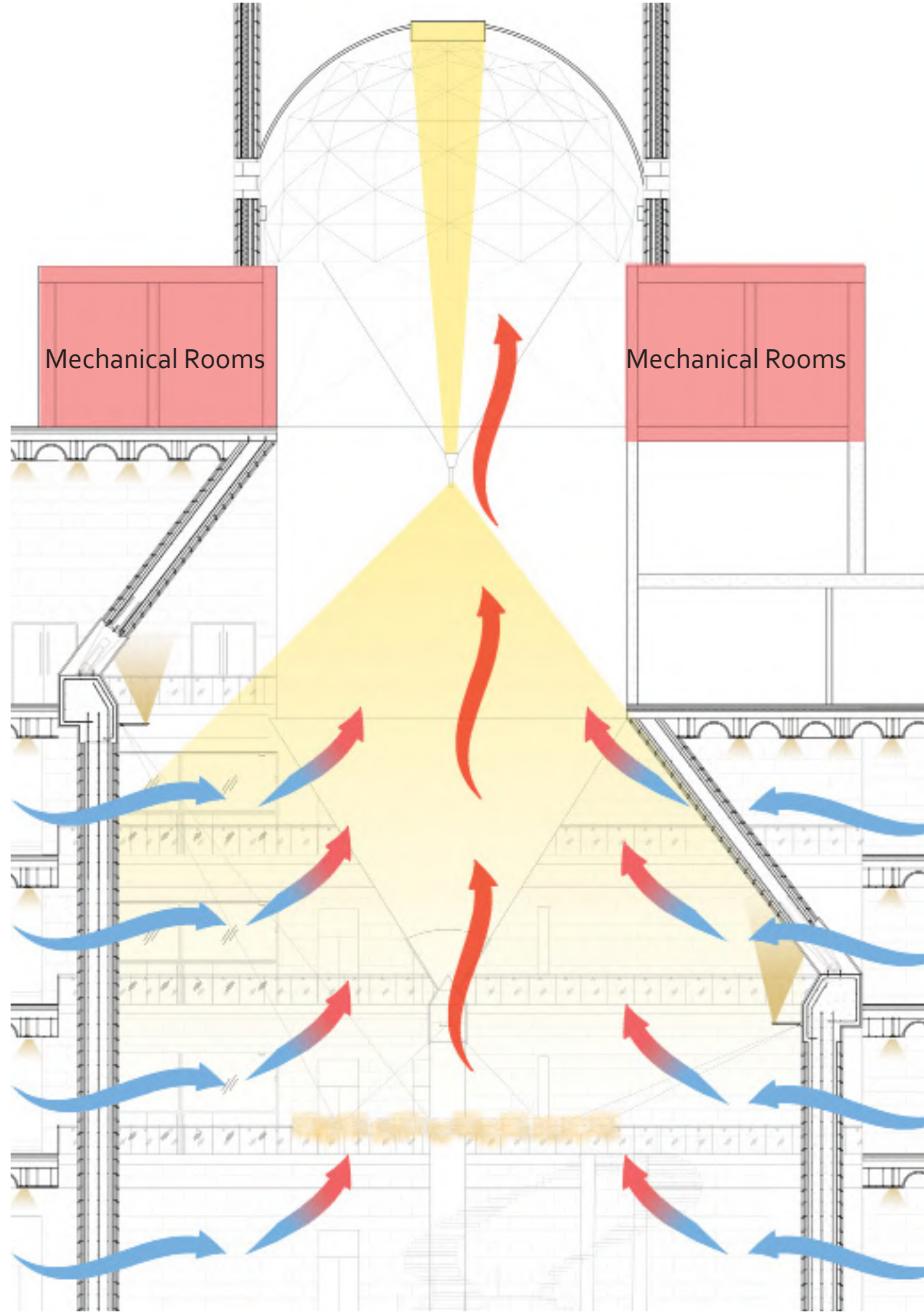
Emerging as a primary example of sustainable and cultural architecture in the middle east, by reaching a carbon neutral, net zero, certification for its operation and services. Making it the first museum in the middle east and asia to reach this goal. The low carbon strategy aligns with with international standards. Positioning it at the forefront of sustainable museum practice and reinforces the role of cultural institutions in advancing broader climate goals



Environmental Analysis



It is important in a hot climate the use of high thermal mass that heat gain and loss was kept to a minimum with the use of thick walls. The use of shade was very important as it is a key element to combat the harsh southern sunlight. Water features are used throughout the project, creating a cooling and calming quality to the space and would work with the incoming winds. The color of the cladding is light which creates a link of the desert of Qatar and has reflective properties so less heat will be absorbed and the air gap between the cladding will add to the insulating properties. The 5 story atrium that overlook the skyline uses double glazed glass and has adjustable screen.



To avoid the disruption of the aesthetic appeal, the mechanical rooms are located at the upper floors of the building. As the feature ceiling runs throughout the building. The HVAC systems and shafts were fitted behind the walls rather than ceilings and AC vents are slits in the wall to make it discreet.



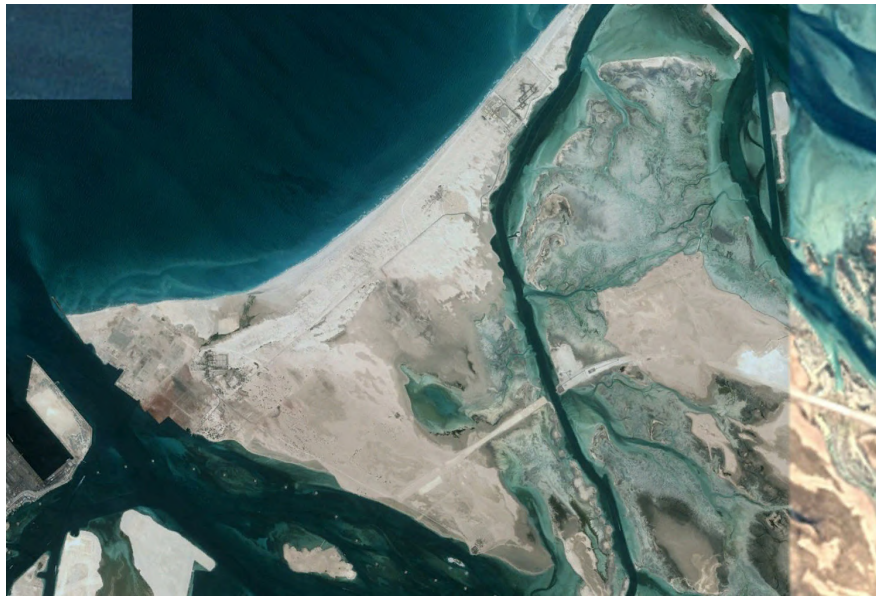
Site Analysis - Cultural District - Saadiyat Island, Abu Dhabi, UAE

The Cultural District on Saadiyat Island in Abu Dhabi is one of the most important nodes in the world representing culture. Bringing together famous institutions that represent both global and regional identity. The Louvre Abu Dhabi, the Guggenheim, Natural History Museum, TeamLab, and the Zayed National Museum are all famous projects that were designed by world famous architects. The district wants to make Abu Dhabi a top place for cultural exchange and creative innovation by holding world class exhibitions, education programs, and interactive public spaces that celebrate art, history, interaction, and culture.

Historical Site Development



1987



2004



2025



Primary Visual Landmarks

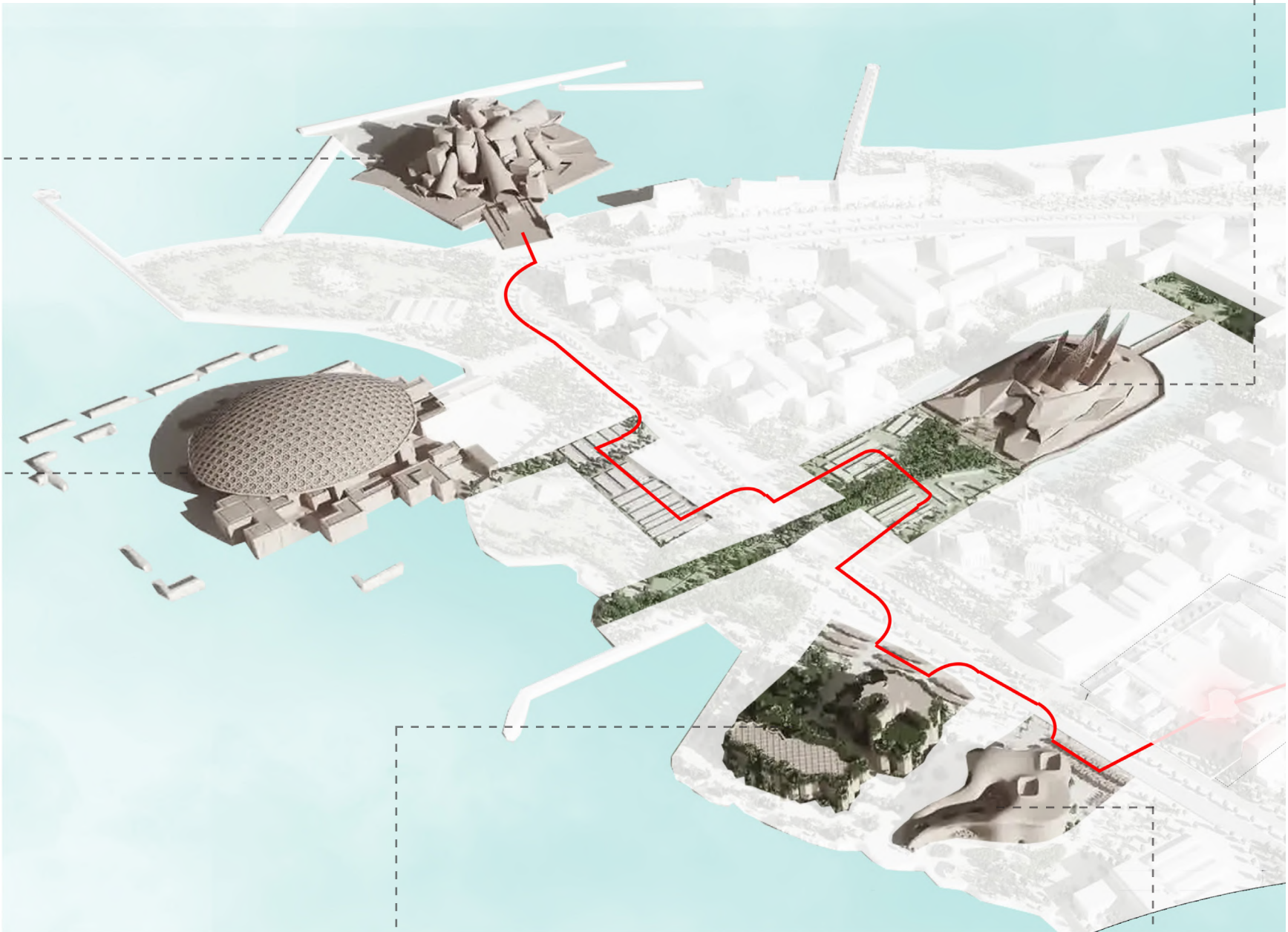
- Guggenheim Museum - Focus on modern & contemporary art from asia.



- Abraham Family House - interfaith dialogue & mutual understanding



- Zayed National Museum - A narrative journey through U.A.E. history



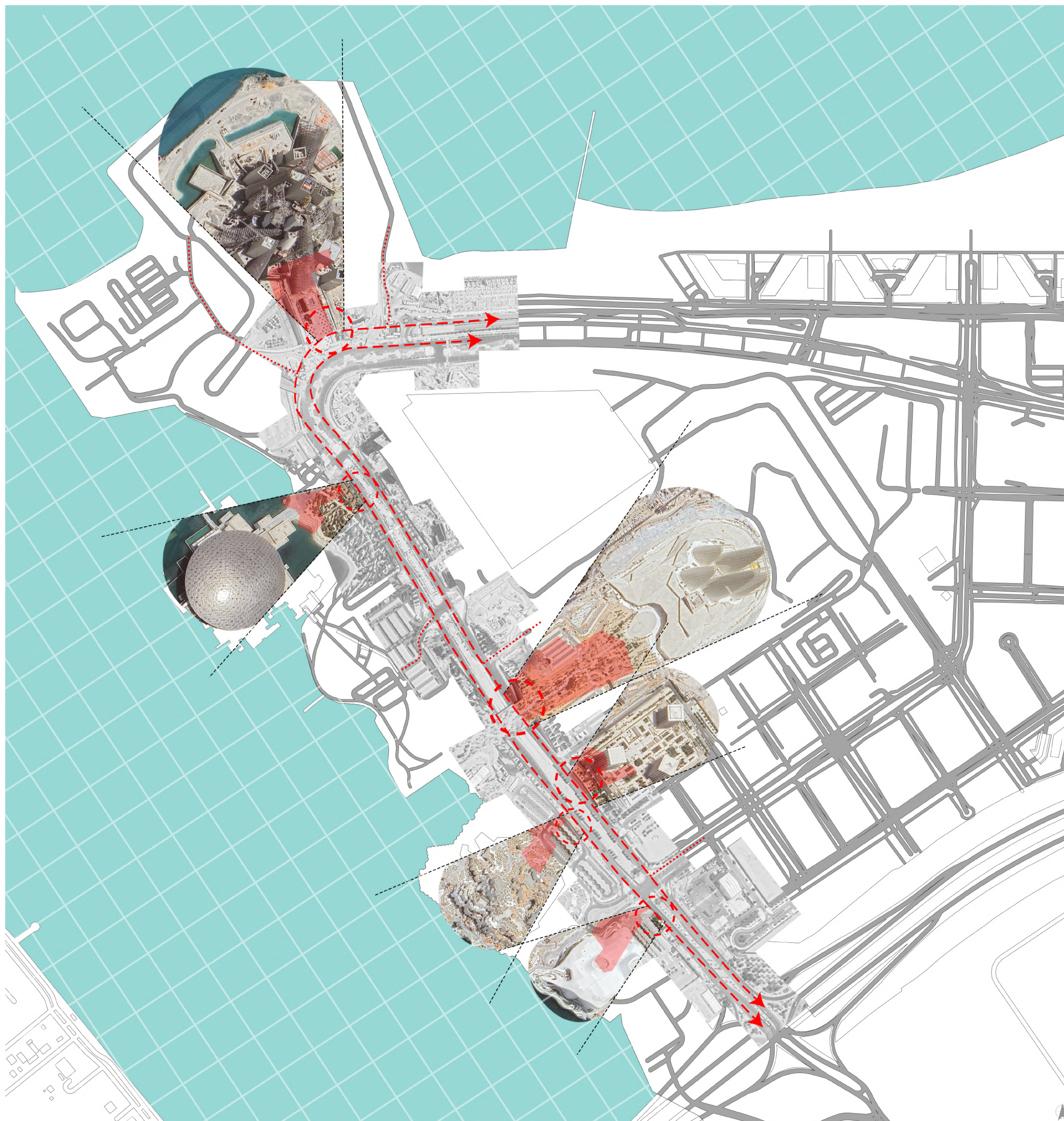
- Louvre - Universal archive, mostle historical art and unique artifacts.



- Natural History Museum - From the beginning of the universe materiality

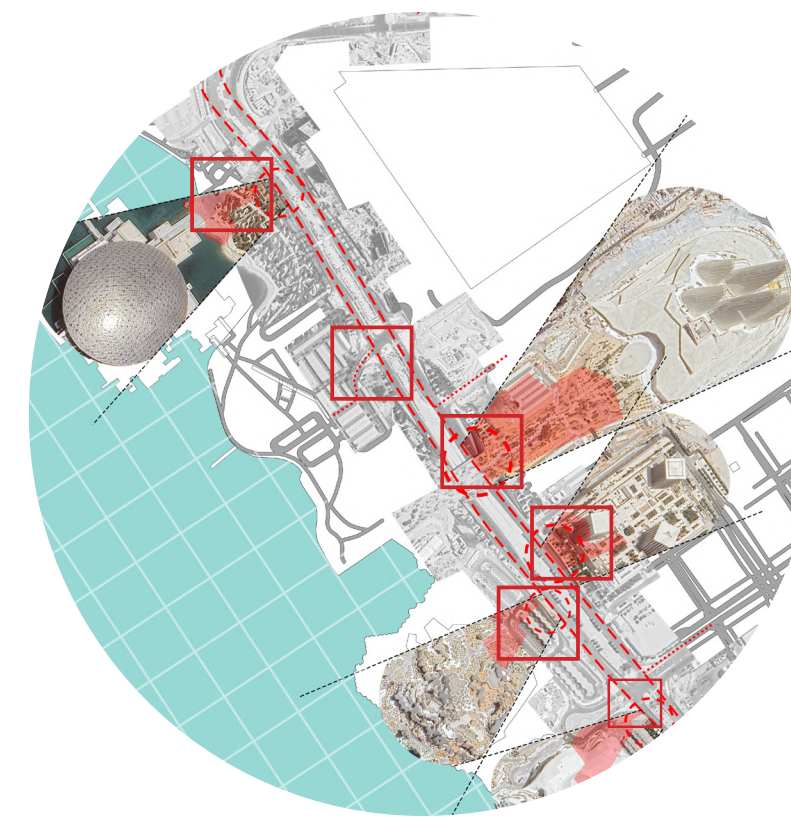


- TeamLab - modern interactive art space



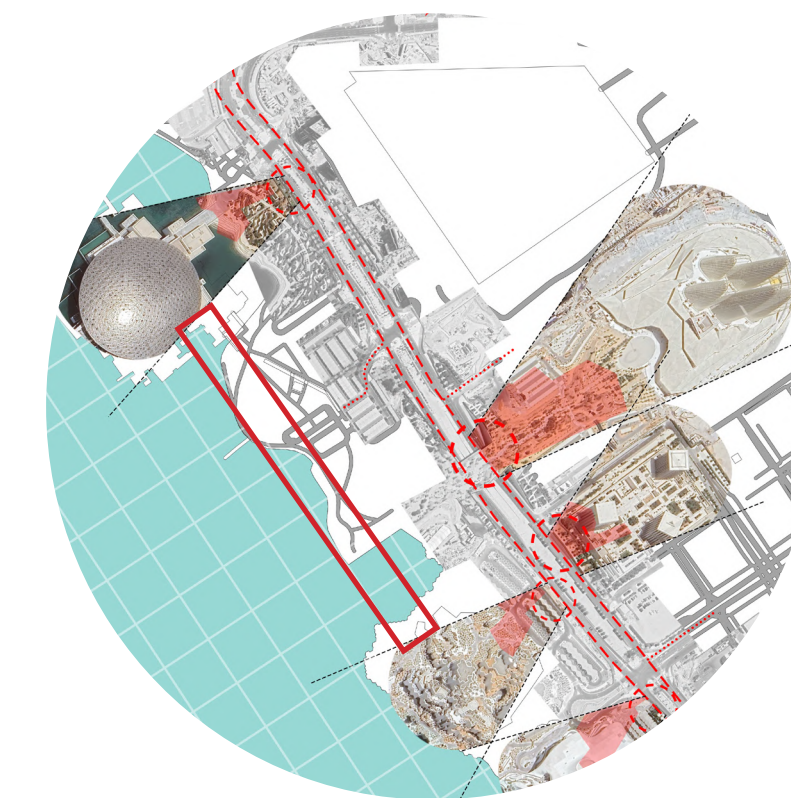
Site Analysis - Linear Strategy

The linear spine is strategically positioned between major cultural attractions in Saadiyat Cultural District, connecting the Louvre Abu Dhabi, teamLab, and the Zayed National Museum. Its elongated form encourages a series of public transitions, creating an activated pedestrian corridor that leverages urban permeability, seafront views, and cultural integration.



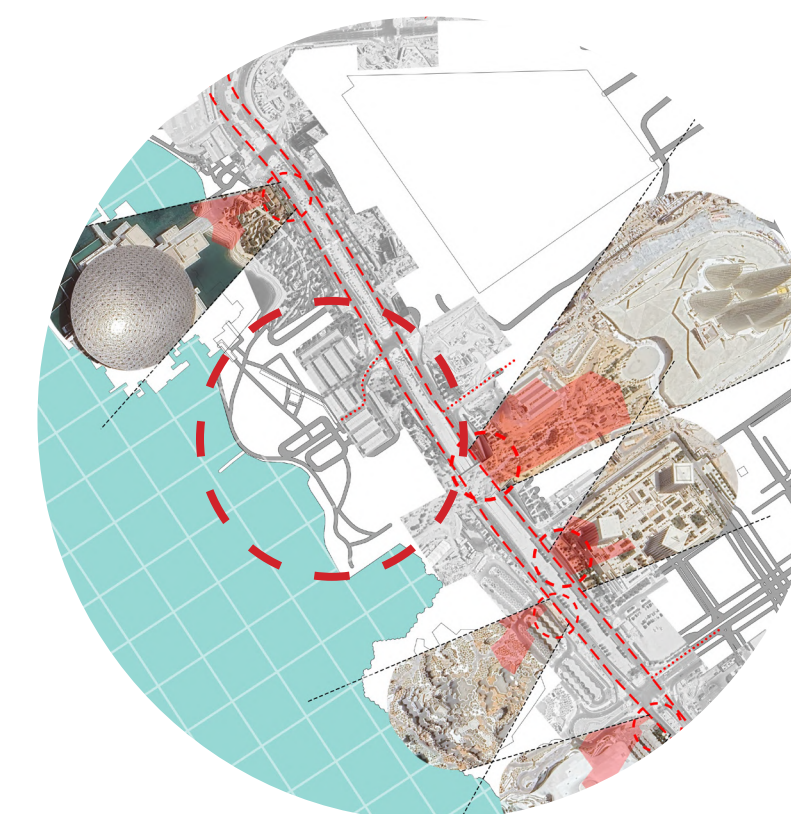
Urban Connection Opportunities

The site functions as a hinge in the district, creating new urban connectivity between cultural institutions. Its proximity to key attractions makes it ideal for linking movement, landscape, and culture.



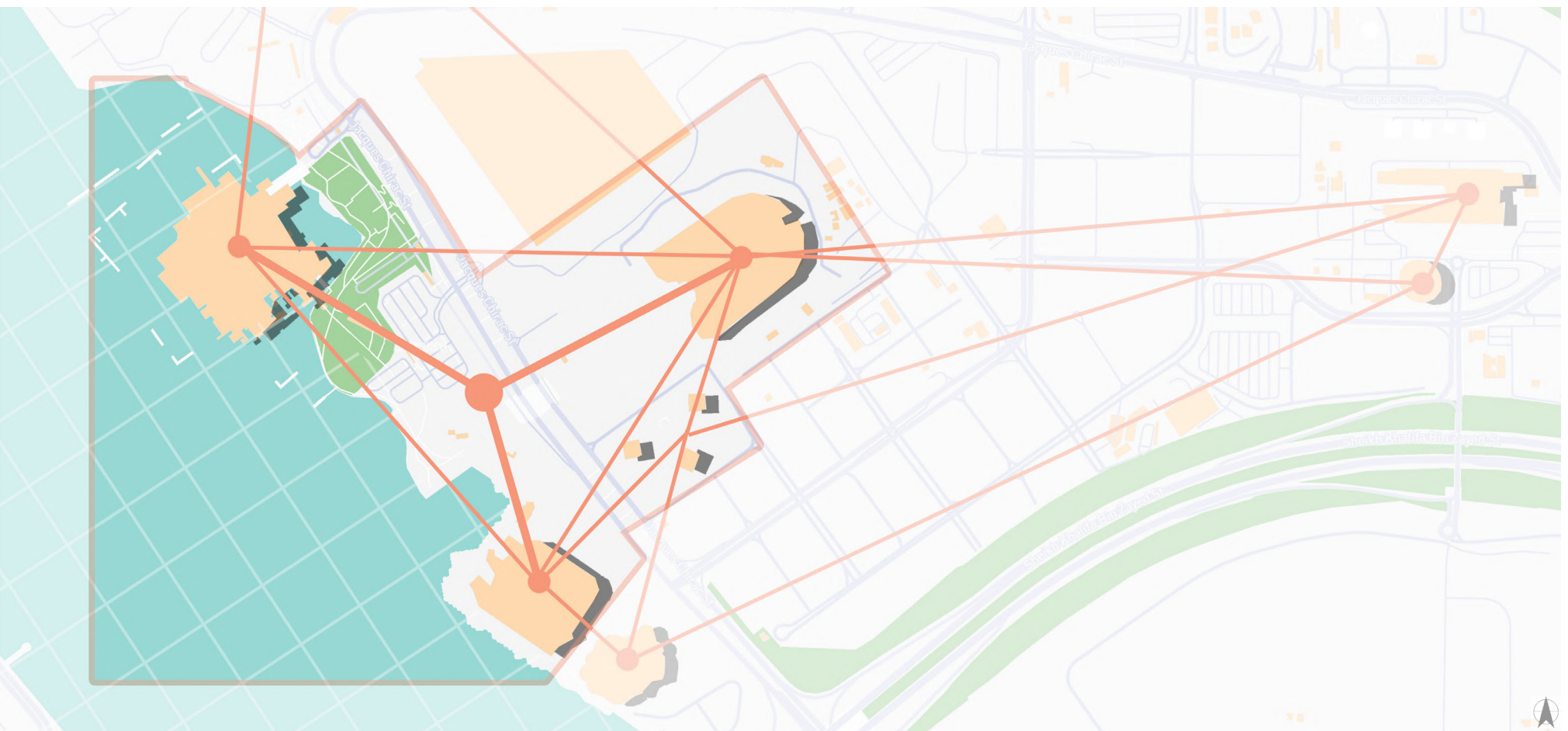
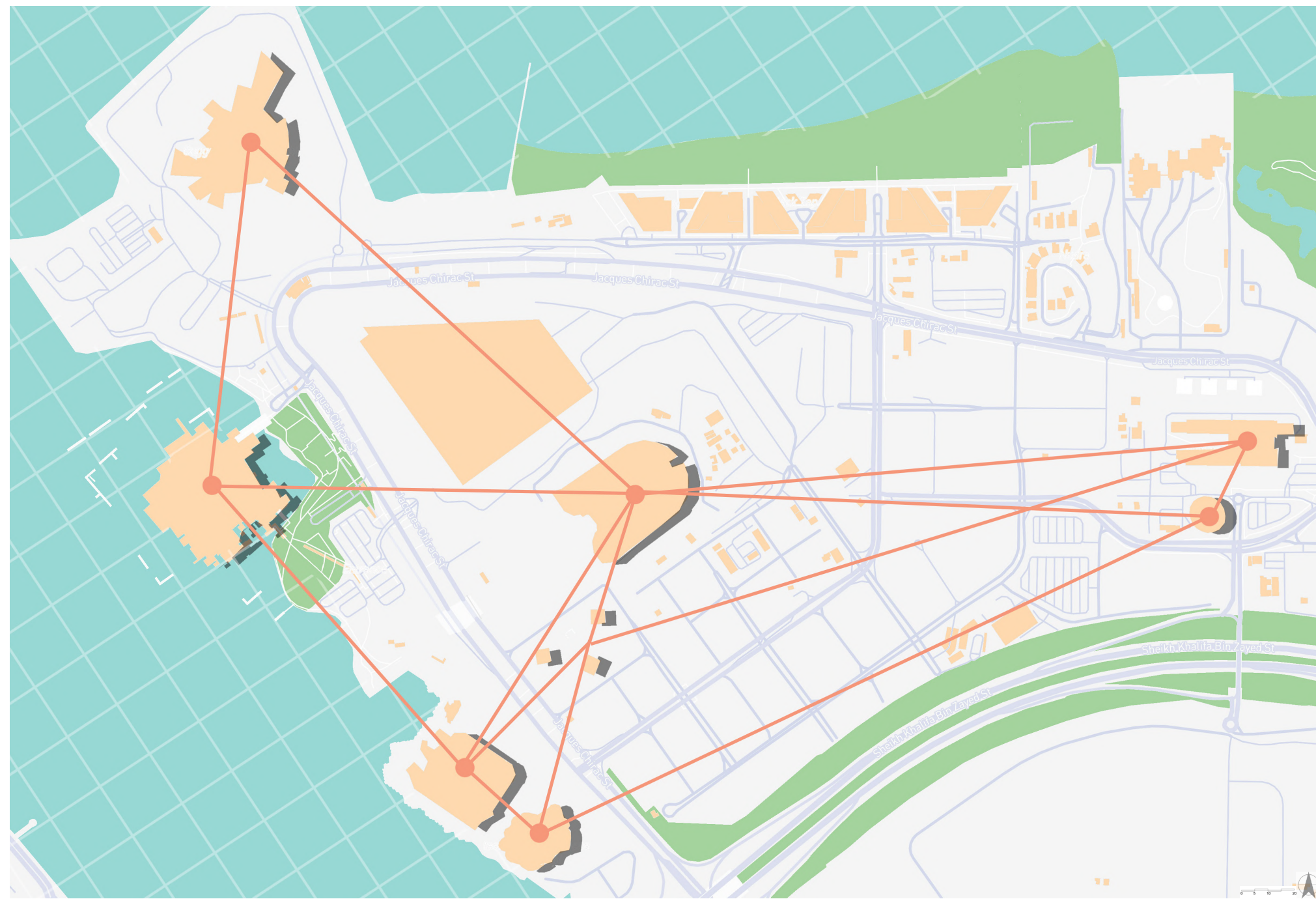
Visual + Spatial Permeability

The corridors opened allows unobstructed views toward the sea, inviting people into a shifting sequence of experiential public spaces.



User - Friendly Walkability

The site's linearity links key landmarks with potential of a walkable route, engaging public flow and activity along an engaging path.



Analysis + Initial Strategy

Positioned strategically within Saadiyat Cultural District, the site has the opportunity to bridge major cultural attractions while addressing environmental and movement factors. An initial design response centers on the creation of a linear cultural corridor that leverages natural exposure to connect institutions while integrating landscape and self-shading strategies

Connectivity — Visibility — Microclimate — Accessibility



Site Selection

The site central location links it to the Louvre and the Zayed National Museum, creating a hinge point for movement across the island.

- Current urban spine under-utilized

Potential for civic reinforcement through program, permeability, and public occupation.

- Direct undisrupted winds from dominant direction

supporting prevailing airflow and passive cooling used to integrate self shading forms.

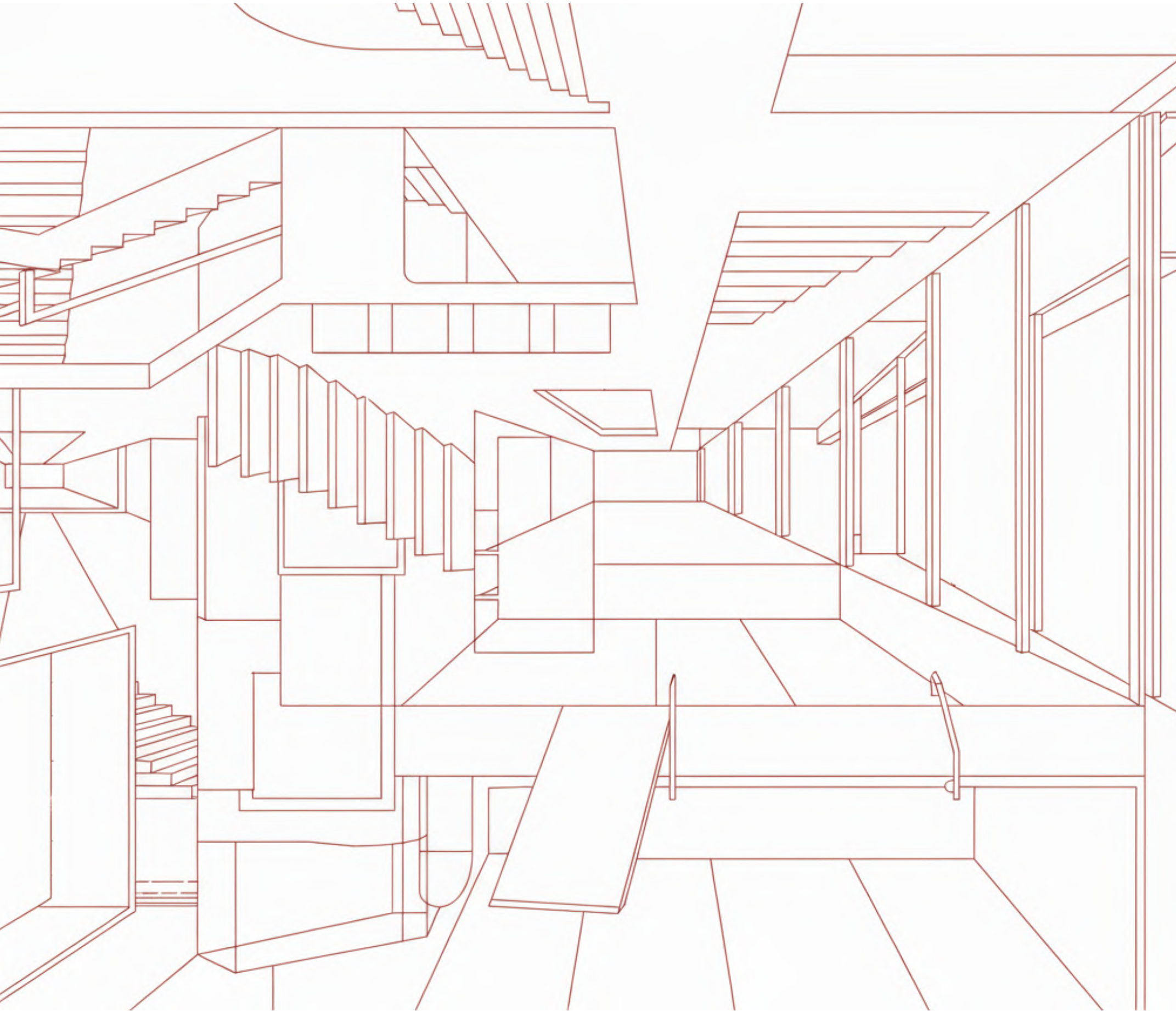
- Site selection location

Close to major cultural destinations to benefit from footfall, yet open to develop its own environmental and spatial identity.

Shape Taxonomy: Order to Disorder

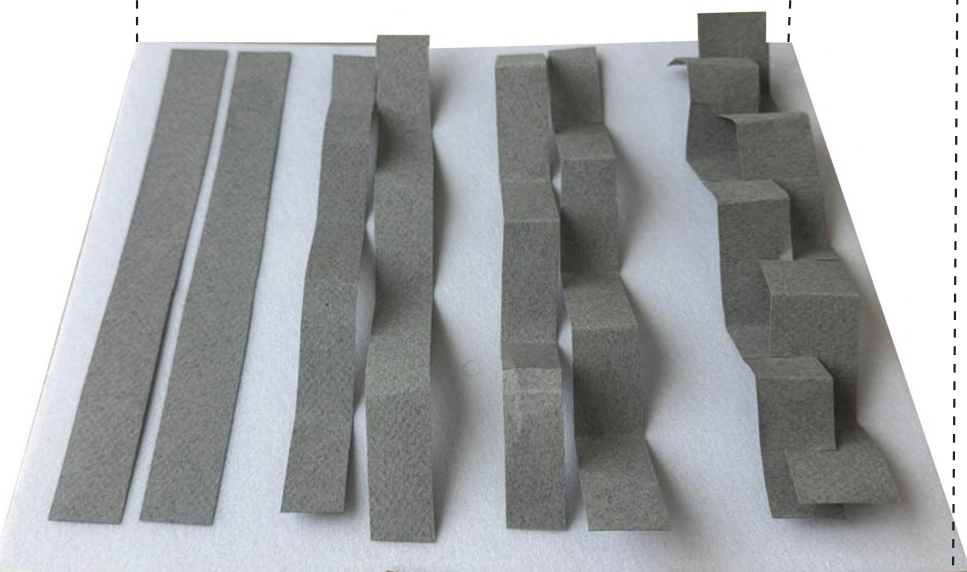
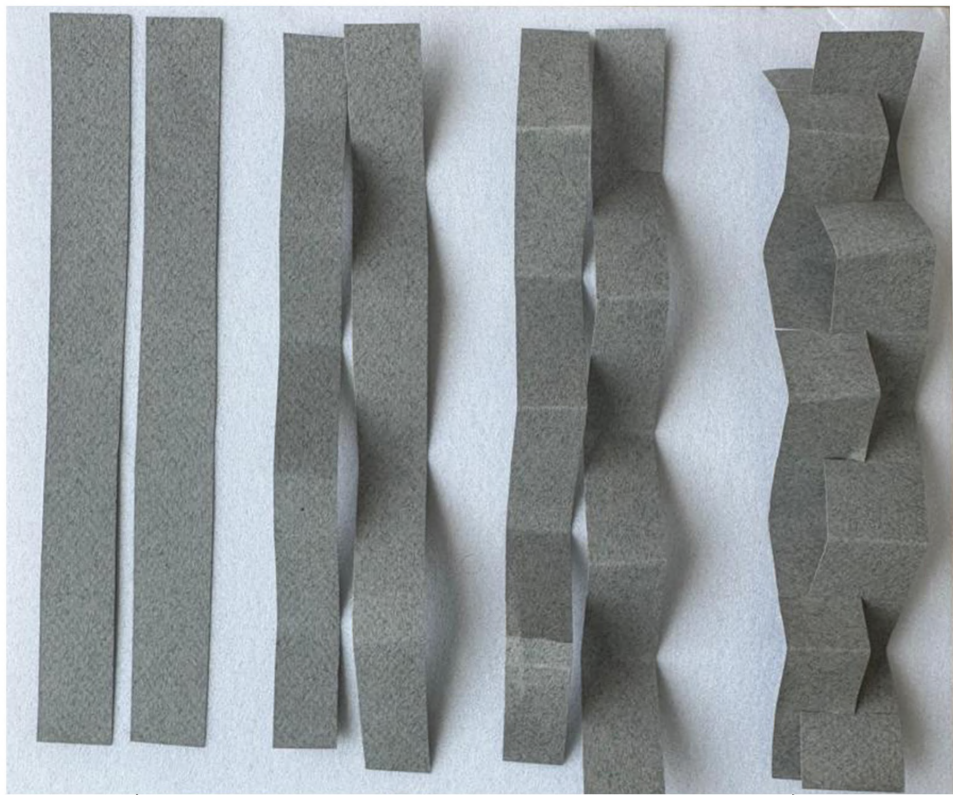
Exploring how ordered geometries can be transformed into dynamic spatial systems through folding, shifting, and distortion. Beginning with linear and controlled forms, the shapes move toward disorder, generating variation, depth, and spatial complexity. Creating moments of overlap, clustering, and transition. Resulting in layered spatial experiences.

Spatial Projection Quality

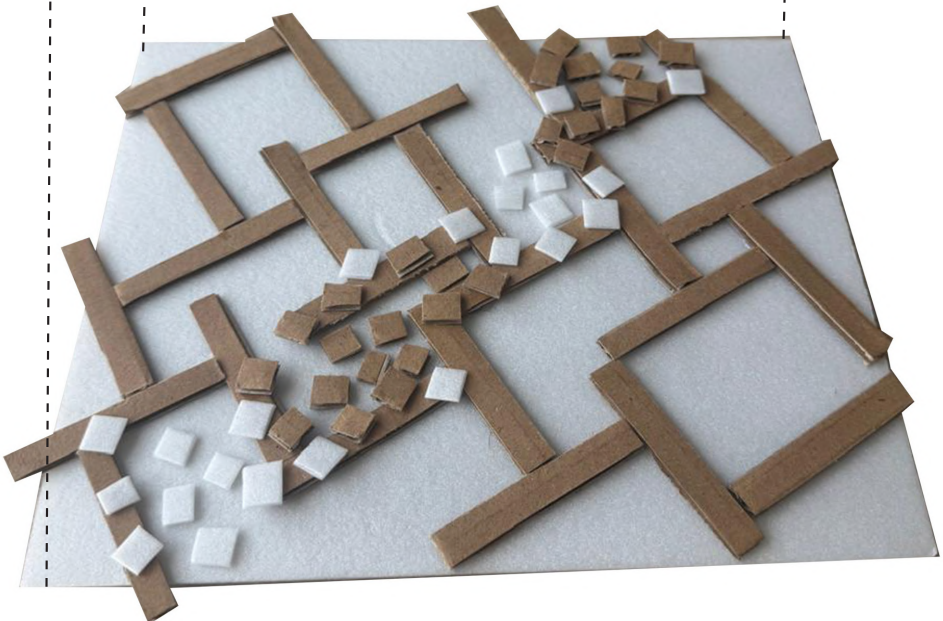
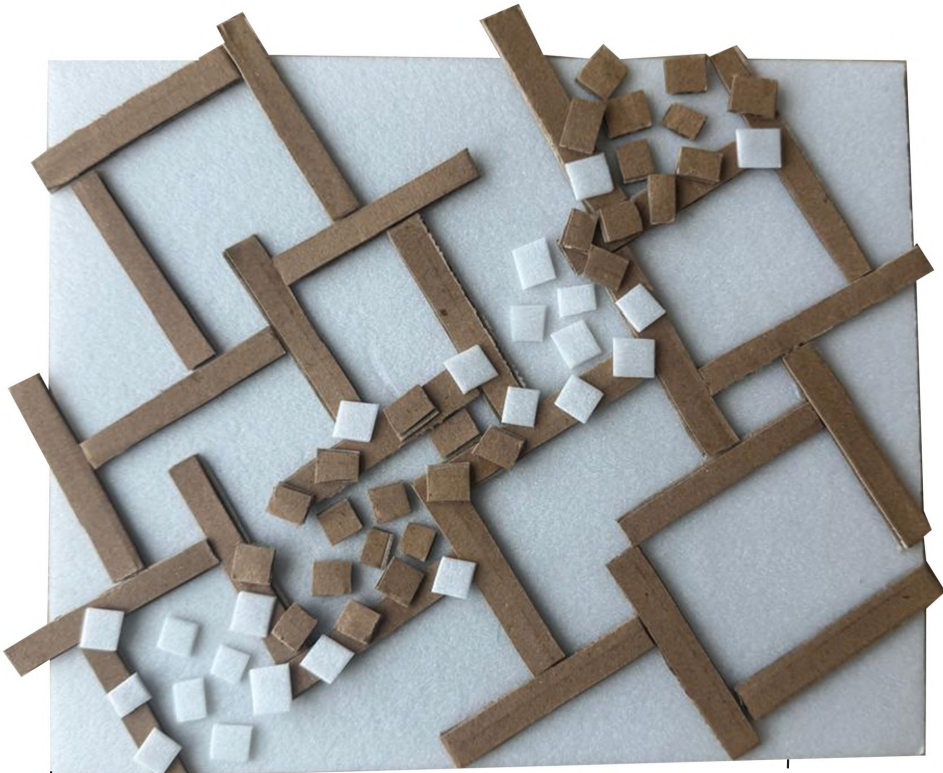
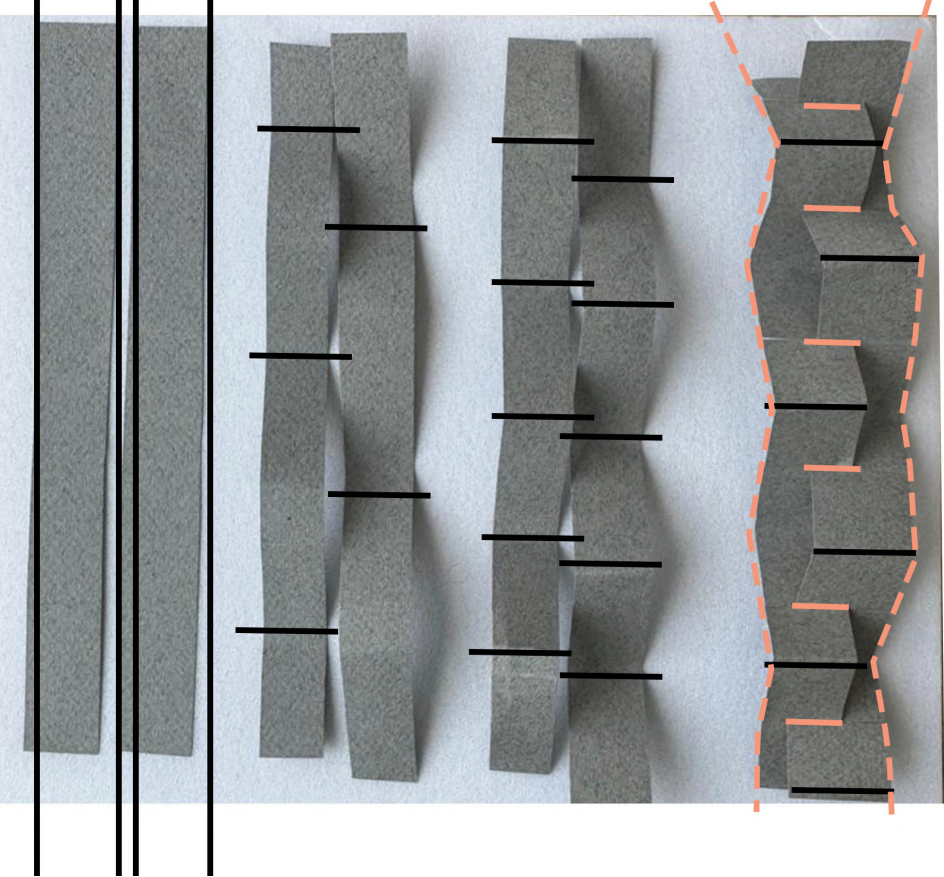


Application to Site Structure

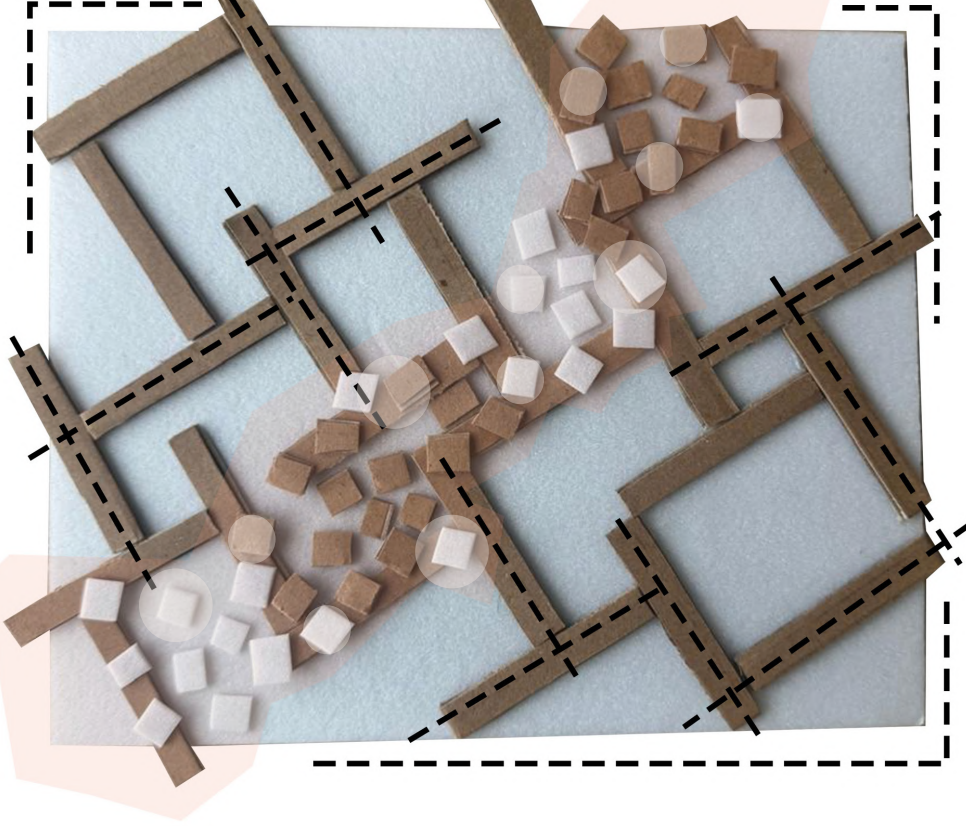
Architecturally, these transformations begin to produce thresholds, framed perspectives, layered circulation, and moments of compression and release based on the site context and future of libraries. The result is a spatial language that is both ordered and dynamic, structured yet adaptable.



Order ————— Disorder



Order ————— Disorder ——— Order



Shape Transformation: Order to Disorder

Through a sequence of manipulations, the strip evolves from parallel order into folded and distorted forms. This process introduces depth, orientation, enclosure, and hierarchy.

Taxonomy of Shapes

Design Development

The taxonomy of shapes is not only a formal study but a spatial system. Triangular and folded geometries are assigned according to patterns of human interaction - person to person, person to knowledge, person to self, and person to nature - while also responding to site conditions and generating. self-shading as an environmental strategy

Person to Person

Folded Planes

Forming shaded canopies
Define gathering pods

Person to Knowledge

Triangular Modules

Creating conversation
Knowledge towers

Person to Self

Inward Folds

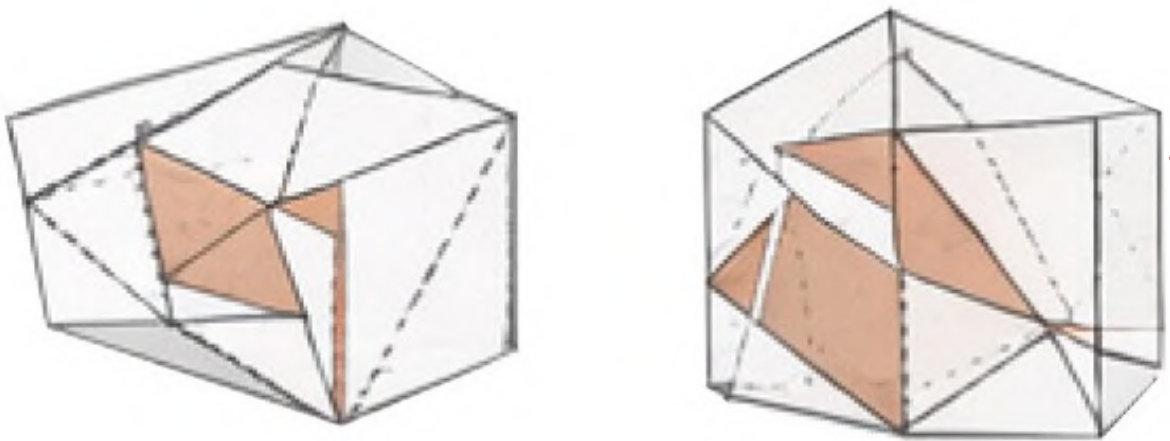
Sensory experience
Inward retreats

Person to Nature

Landscape Frames

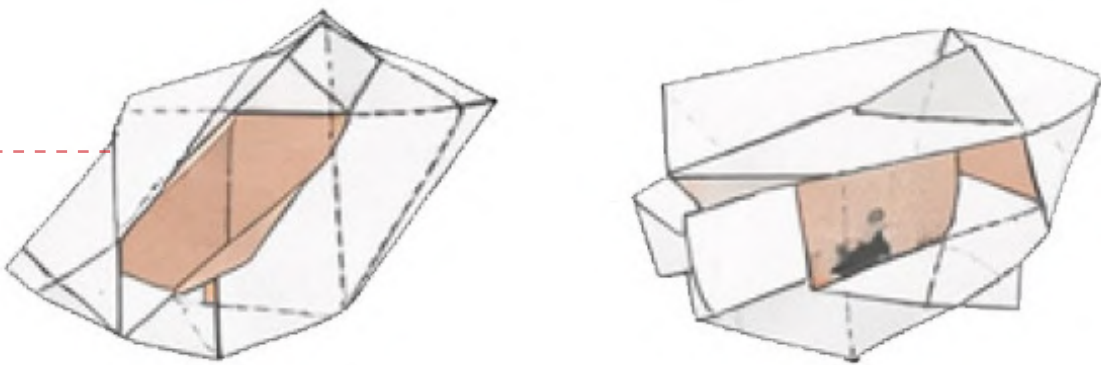
Courtyards
Organic planted pockets

Triangular Modules



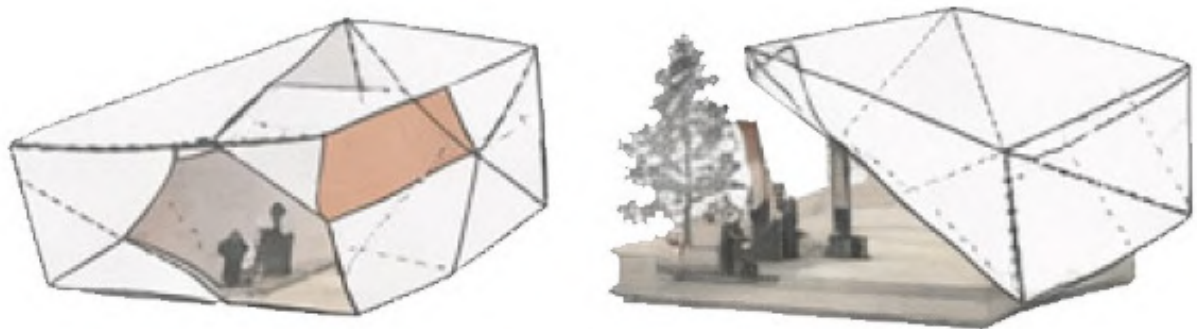
Light wells formed through
cracks of folding created

Inward Folds



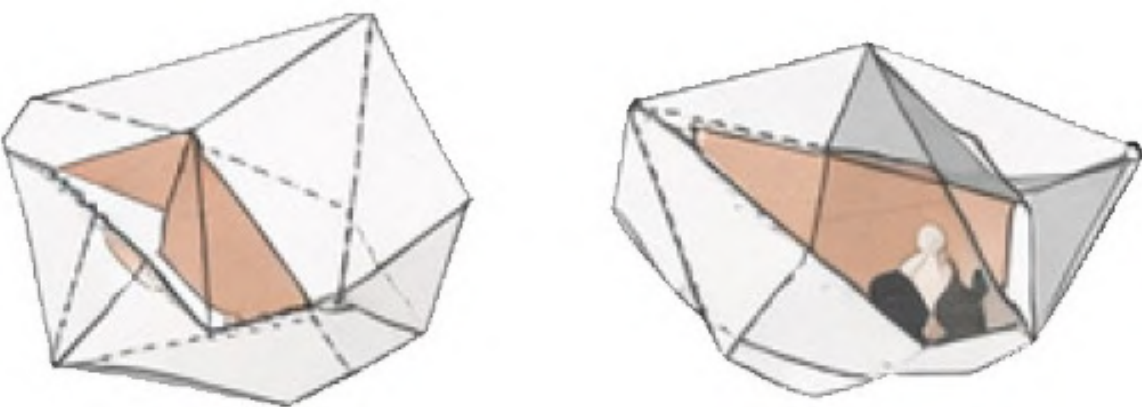
Porous edges allowing reading
hotspots and experience

Landscape Frames

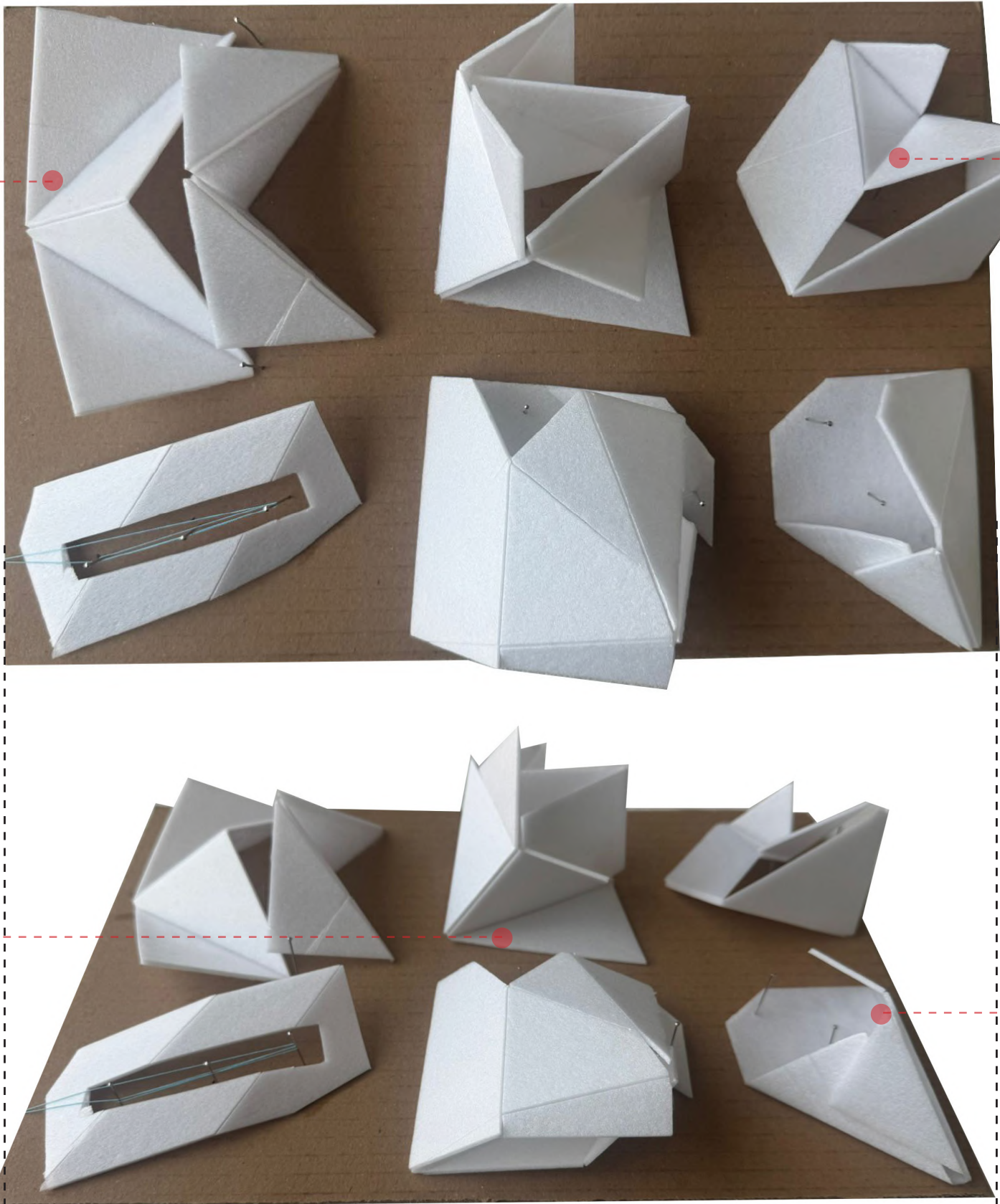


Wind corridors created enabling
threshold transitions

Folded Planes



Defining gathering spaces,
creating an inviting journey



Site Development

Louvre Entrance

Main entry to site

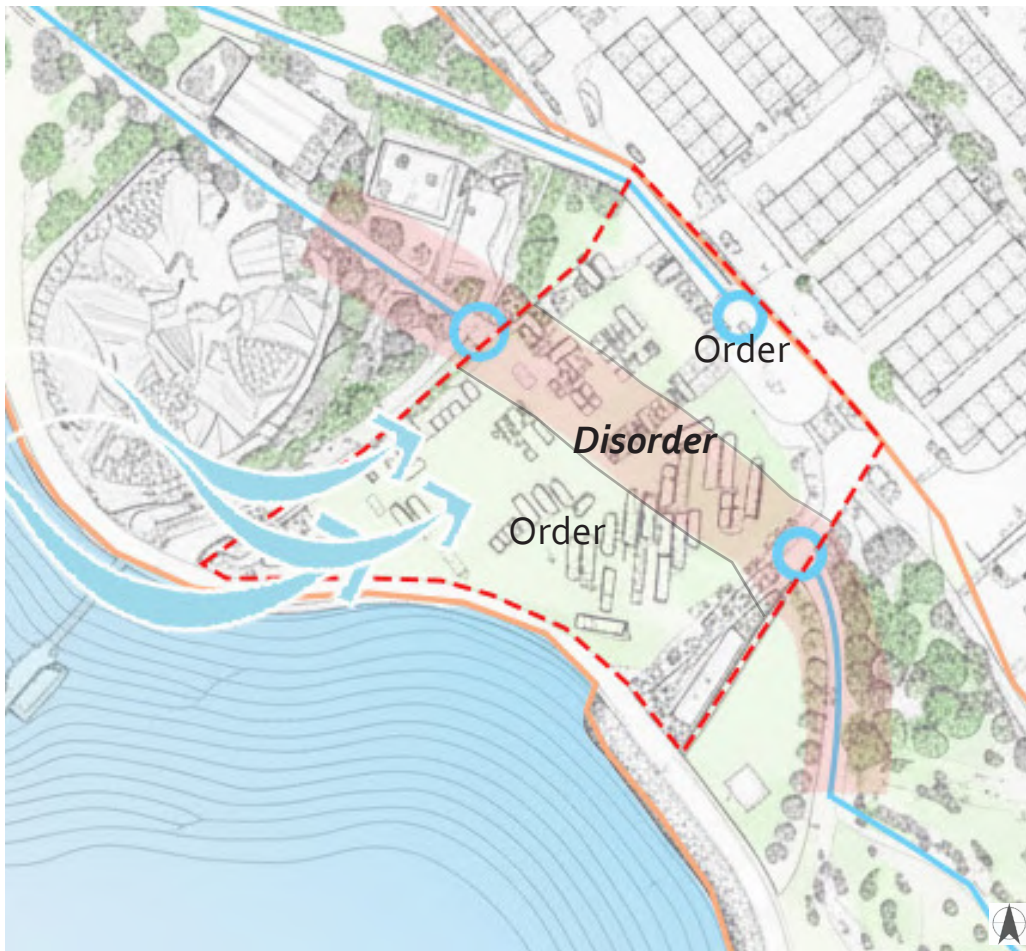
Building spine

Proximity to edge

Primary footpath

Urban Spine

Zayed National Museum



Utilize the wind along the coast

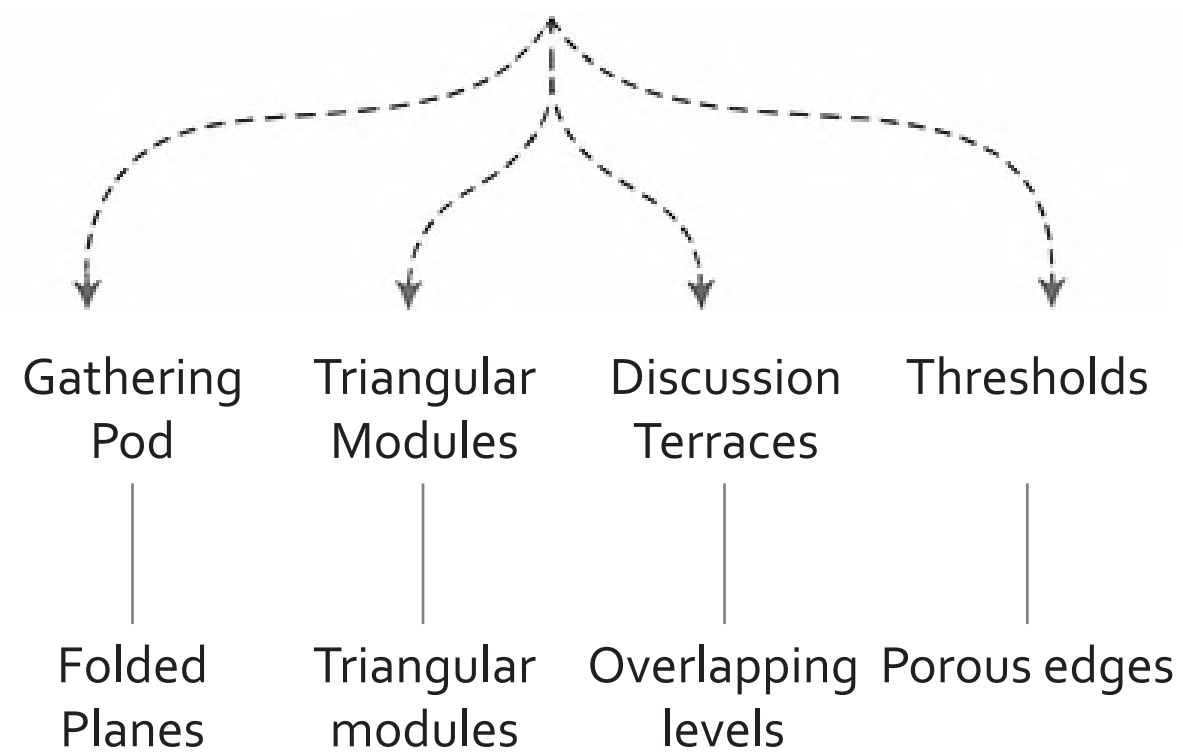


Allowing functions to create a spine

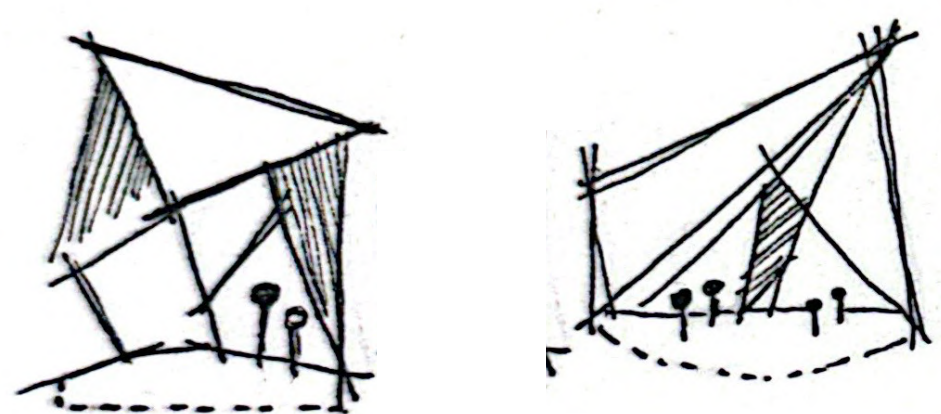


Creating a journey from within

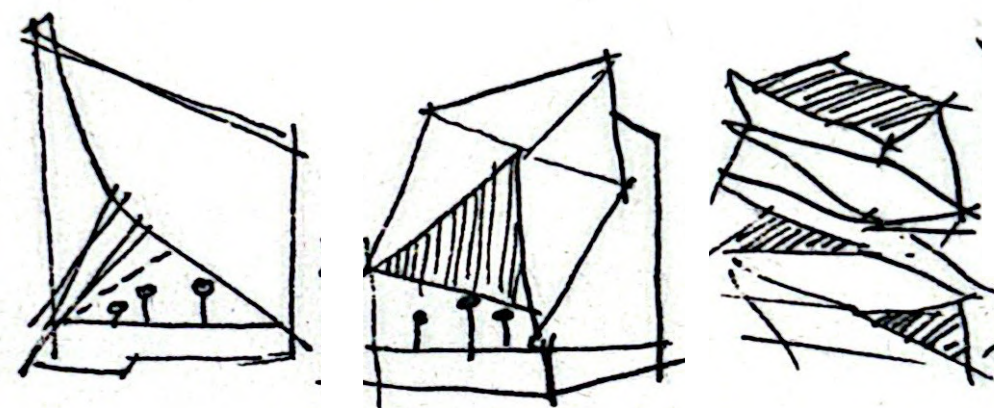
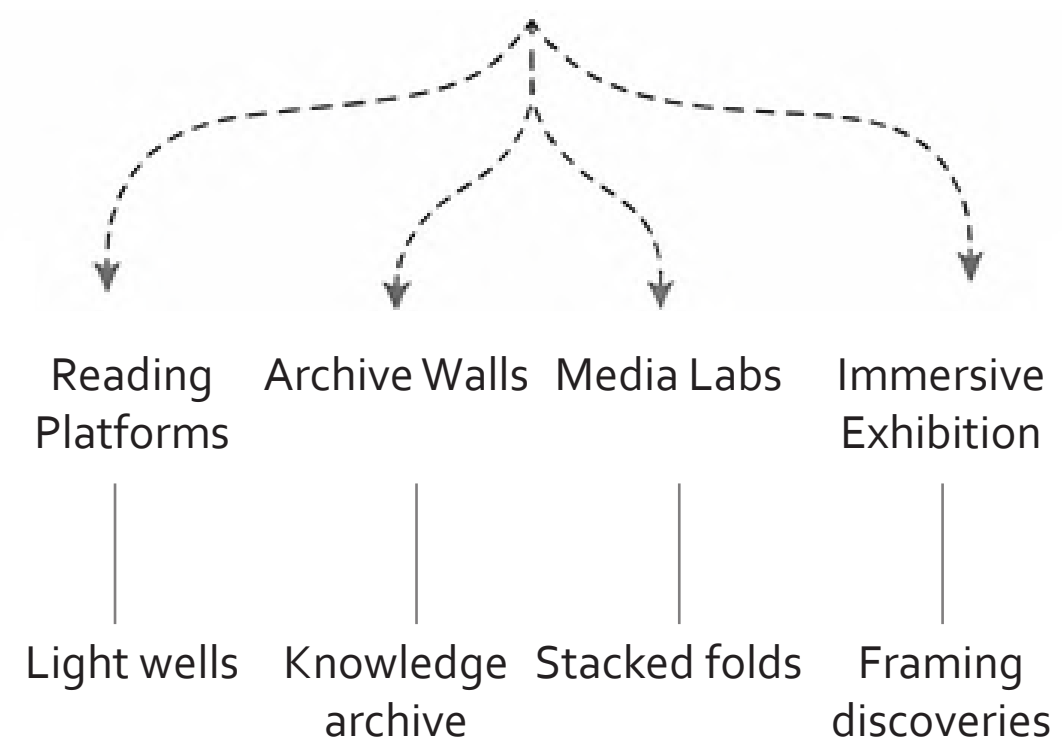
Person to Person



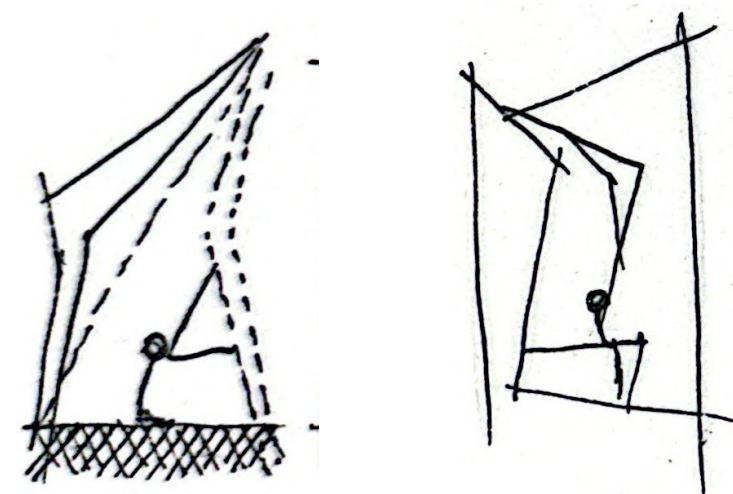
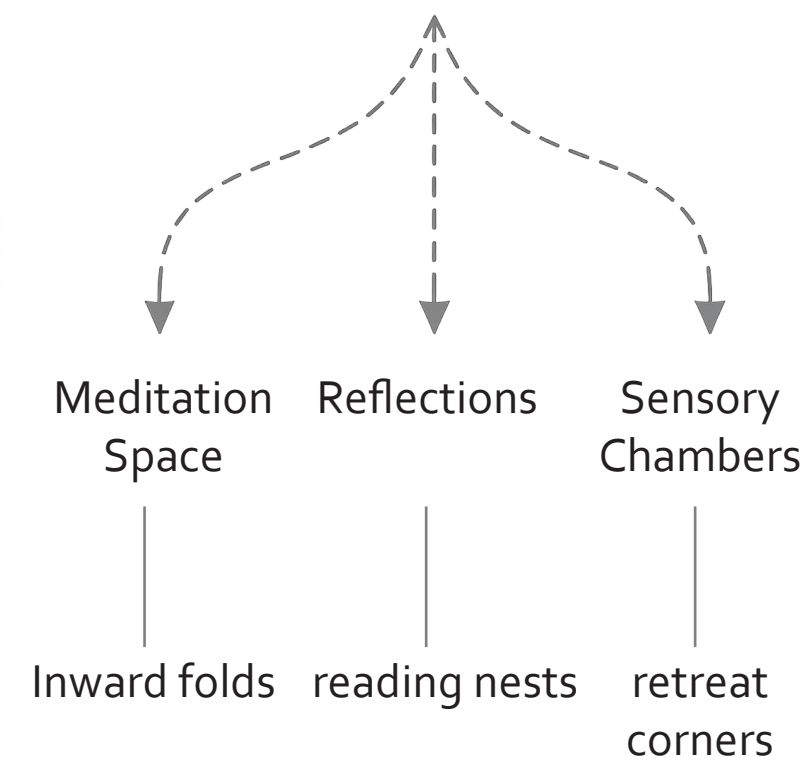
Shape Taxonomy based on program



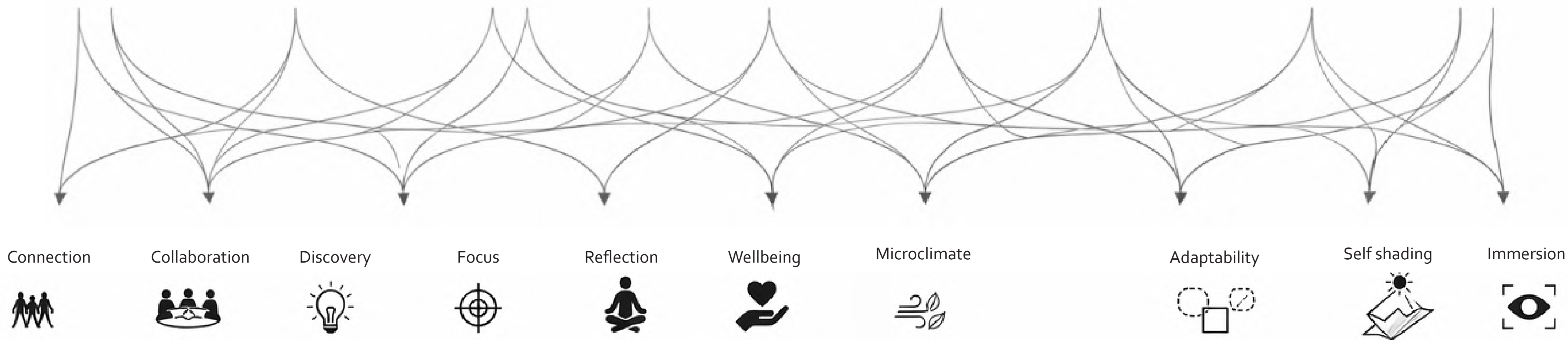
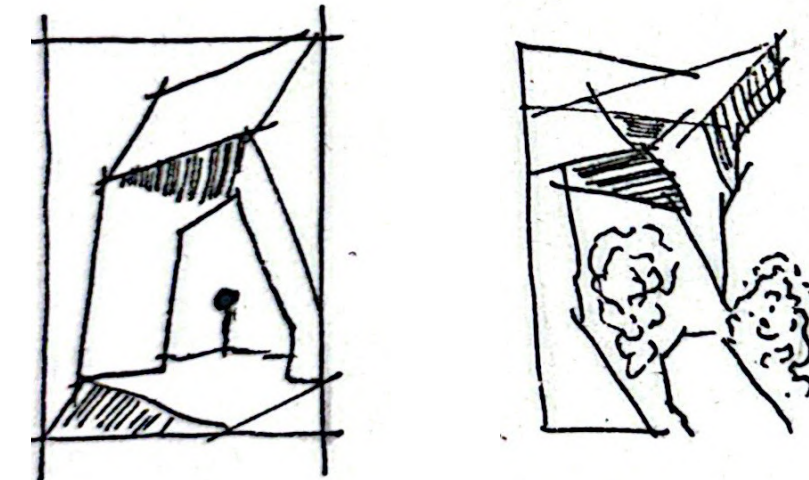
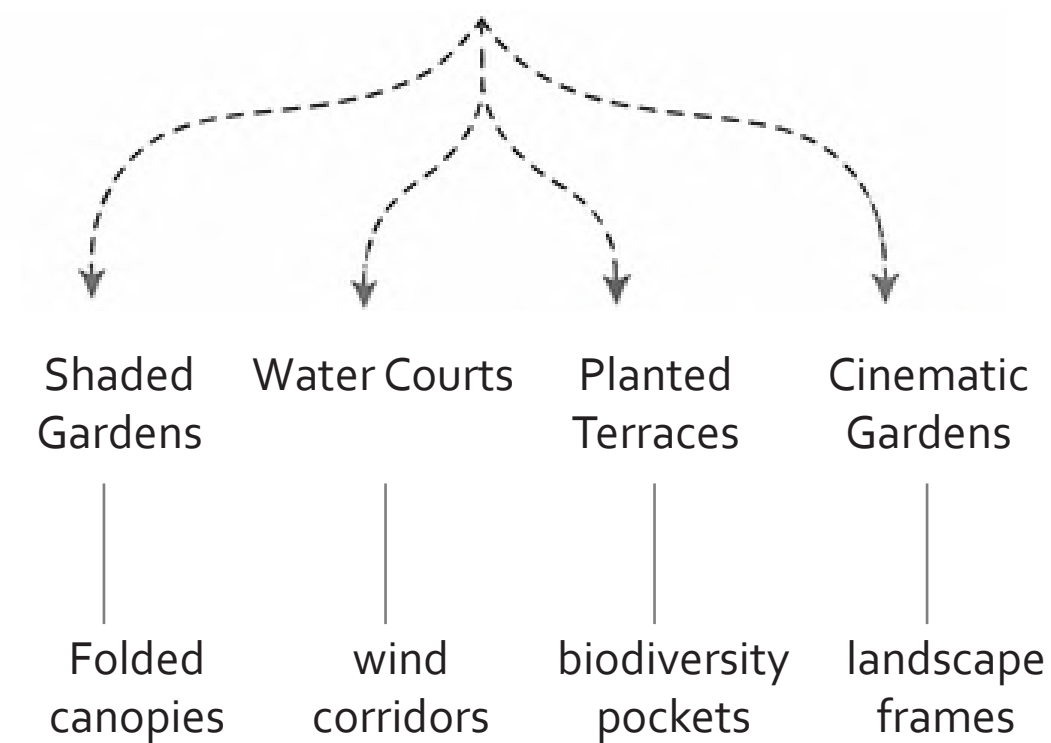
Person to Knowledge



Person to Self



Person to Nature



Developing Spatial Qualities

Site Journey



Implementing person to self, person to knowledge, person to person, person to nature on site.

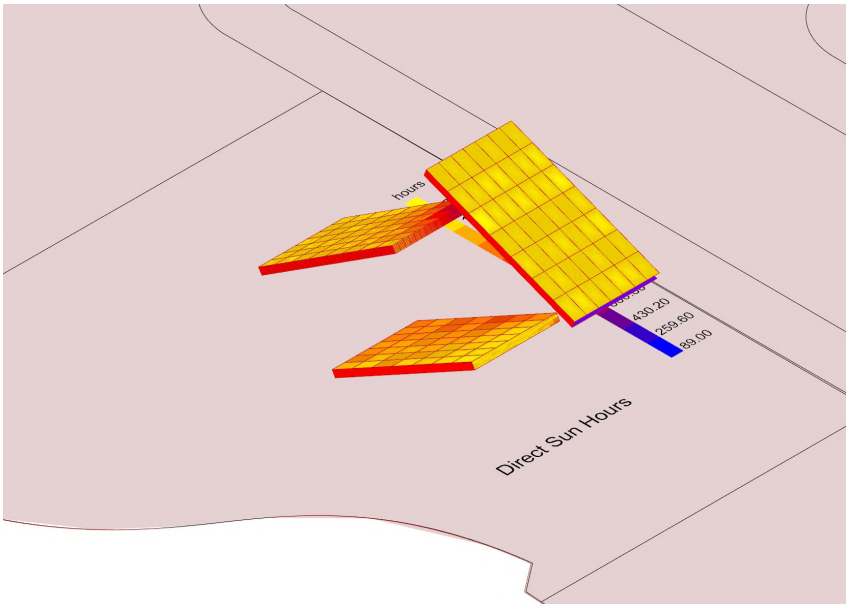
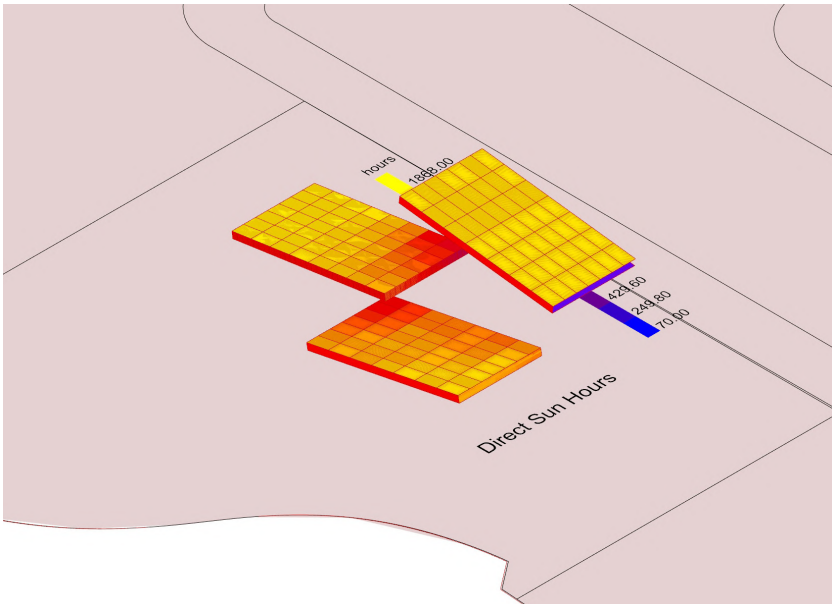
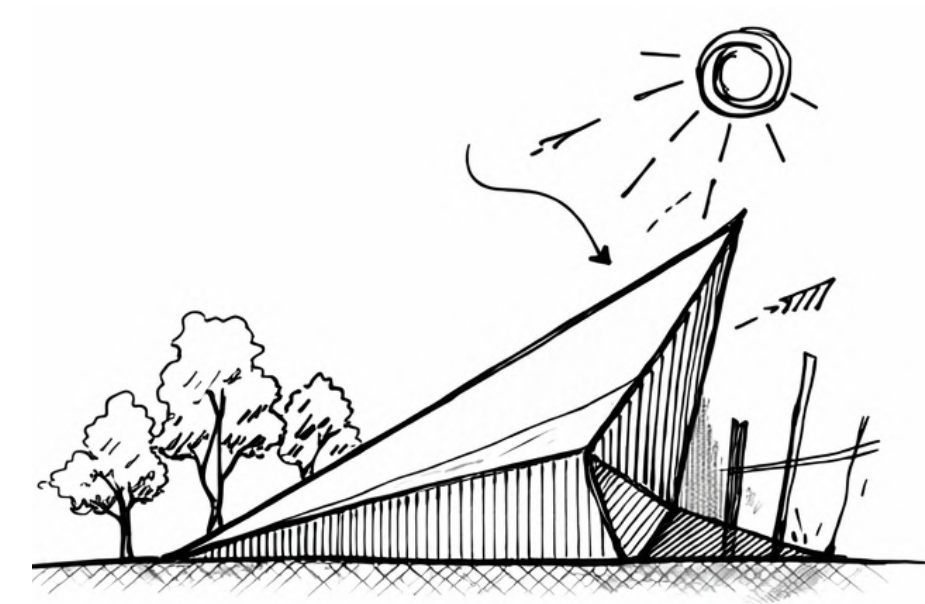


Journey between folds, functions, and experience

Area Program

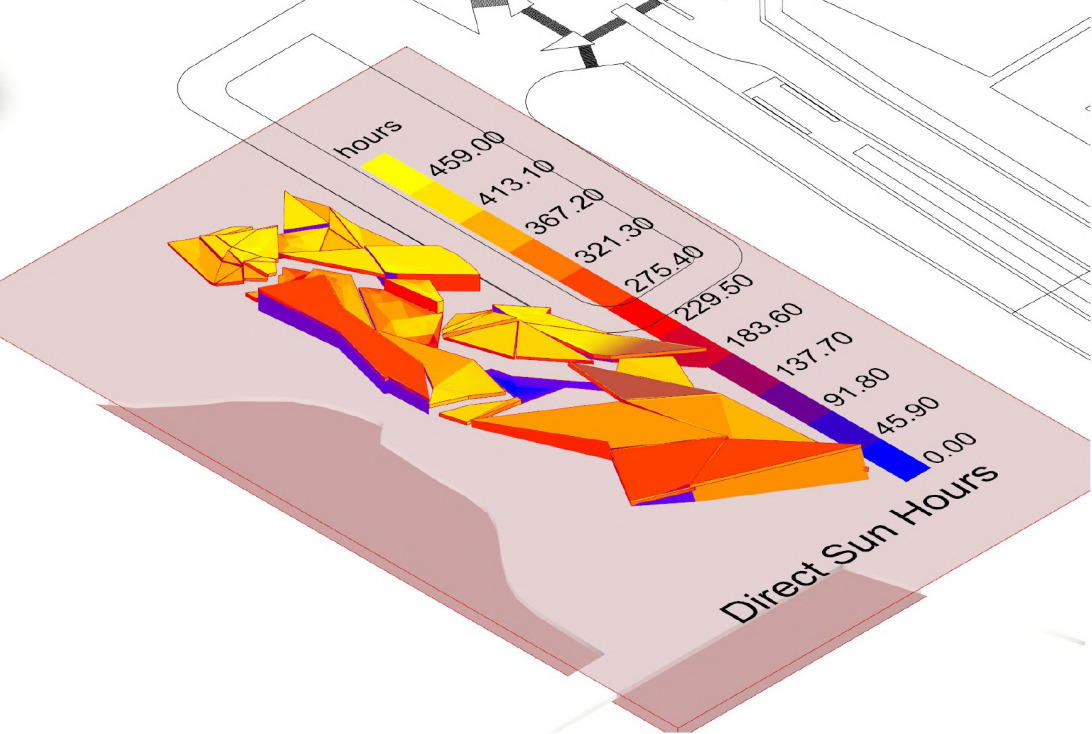
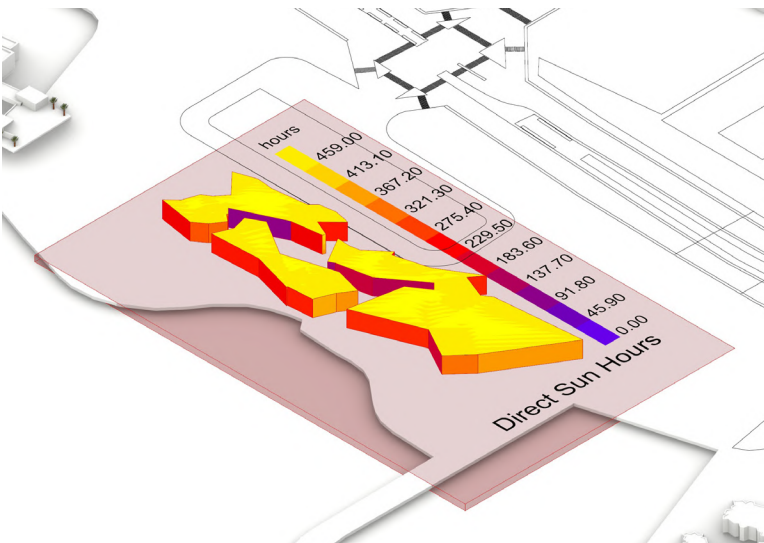
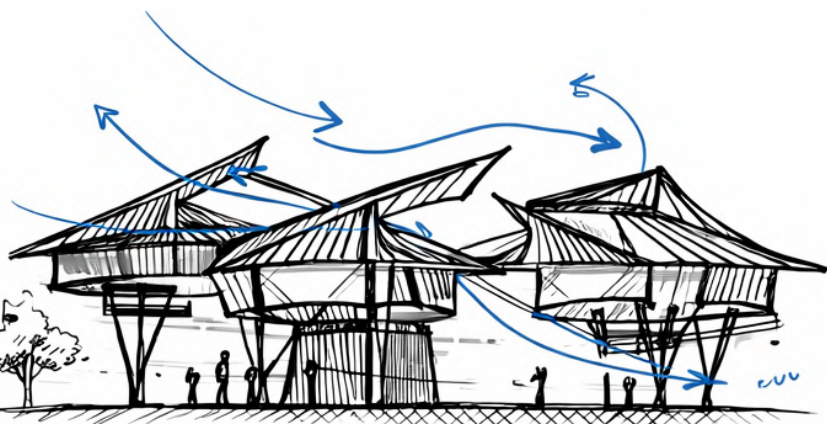
Spaces no.	Functions	SQM	quantity	occupancy	Total SQM
Library Functions					
1	Main reception lobby	140	1	60	140
	Person to Person				
2	Auditorium	700	1	450	700
3	Archive	200	1	120	200
4	Security office	25	1	4	25
5	Administrative offices	15	4	5	60
6	Meeting room	40	4	5	160
7	Youth Lab	110	1	50	110
8	Environment office	150	1	80	150
9	Conference Hall	450	1	200	450
10	Conference Lounge	200	1	200	200
	Person to Knowledge				
11	Coworking space	900	1	600	900
12	Temporary exhibition - sustainability and social development space	700	1	450	700
13	Educational collaboration space	900	1	600	900
14	Exhibition - Emerging Landscapes	580	1	200	580
15	Exhibition - Immersive Canopies of Nature	580	1	200	580
	Person to Nature				
16	Cinematic garden journey	600	1	250	600
17	Biodiversity experience	750	1	450	750
18	Sunken gardens of nature	1200	1	900	1200
	Person to self				
19	Computer lab	135	1	20	135
20	retreat corner space	100	1	120	200
21	sensory chamber	135	1	20	135
22	Meeting rooms	15	7	3	105
23	Study pods	15	4	3	60
24	Admin offices	200	2	60	400
25	Security office	140	1	60	140
26	Artifact Storage space	15	1	3	15
		50	2	3	100
Café and Bookshop					
27	Café	360	1	180	360
28	lounge	400	1	300	400
	Kitchen	95	1	50	160
29	Lounge	300	1	110	300
30	Storage	188	1	50	188
31	Bookshop	550	2	250	1,100
32	Total				
Services					
33	Core	80	4	320	320
	Staircase				
	Fire escape				
	Lifts				
34	IT room	30	1	2	60
35	Electrical room	15			15
36	Plant room	15			15
37	Maintenance	15			15
38	Corridors				
Total Area:					12,628m2

Initial Site Solar Study

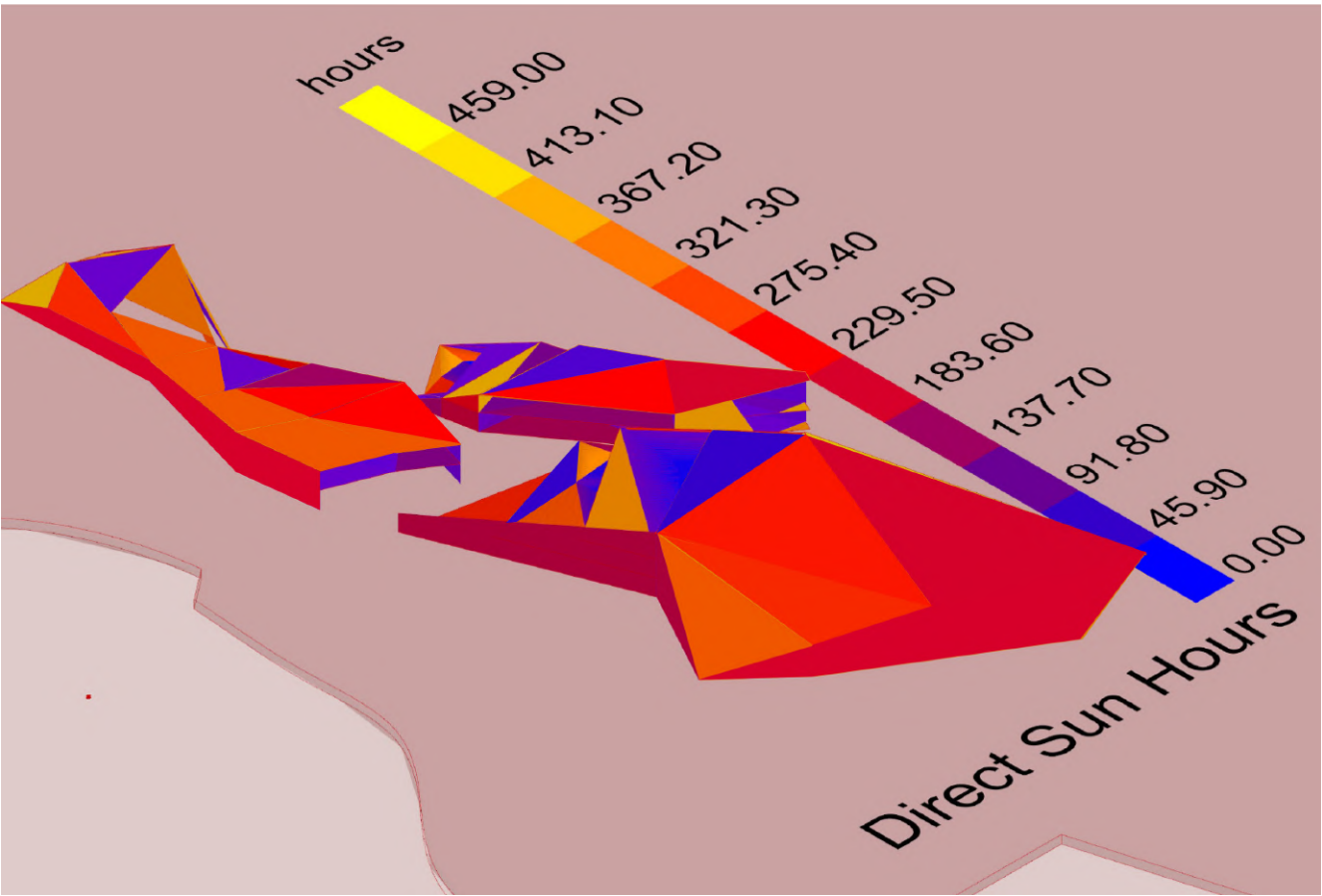


An initial solar study was conducted to optimize the orientation of the building masses for natural daylighting and shape while not allowing harsh direct heat gain within the interior spatial quality. The analysis informed tapering the edges toward the south reduces the harsh heat gains.

Environmental Elements



Wind catching diagram creating the courtyard in the center of the building study, thus elaborating that the flat roof does not respond to the harsh sunlight, folding the roof strucutre to adapt to the climate is used in the deisgn.

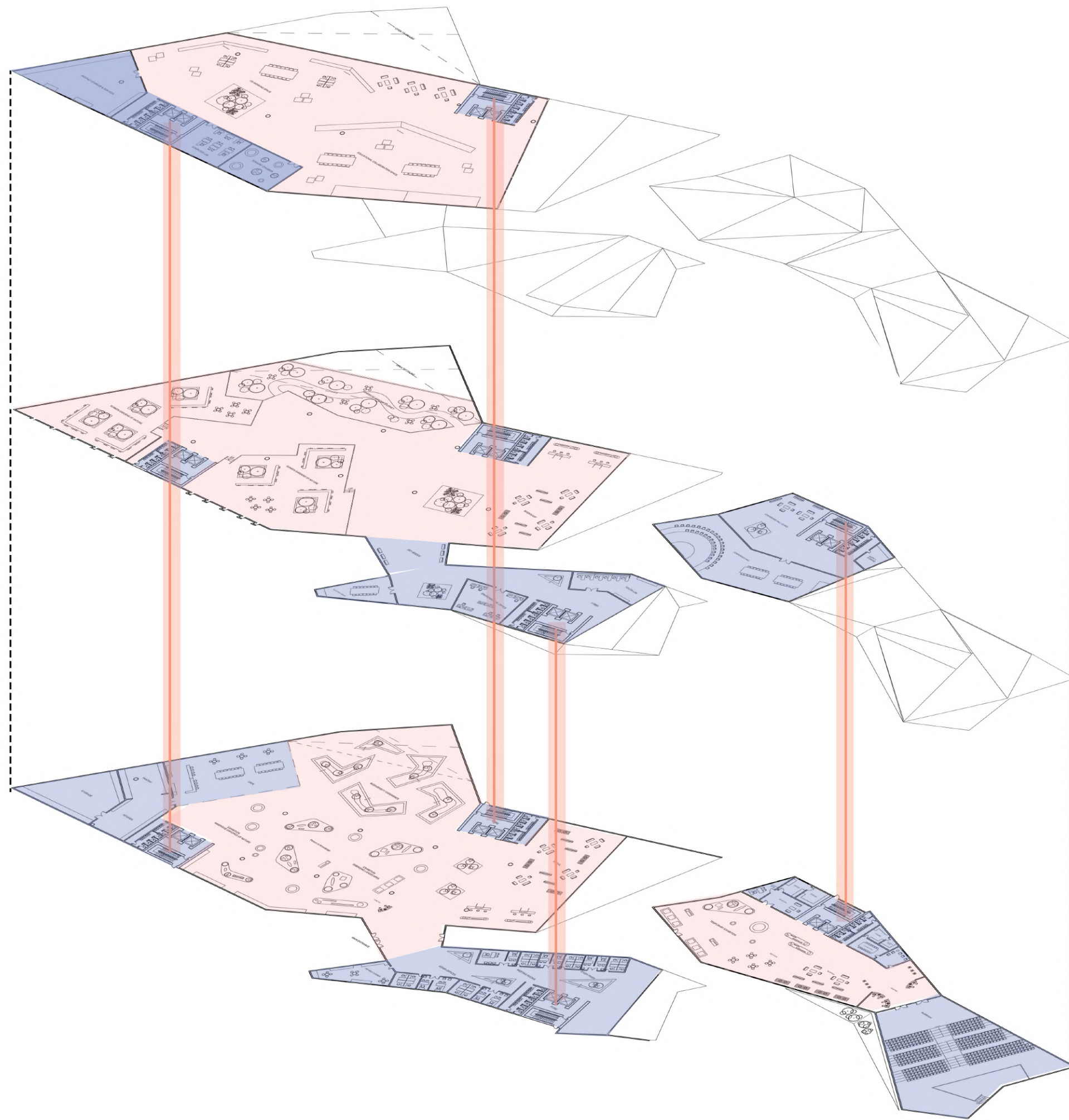


The final environmental study of direct sun on the structure has significantly reduces that harsh heat solar gain on the building thus creating a comfortable environment for the context in the interior and its surrounding.

SITE PLAN

1:1000

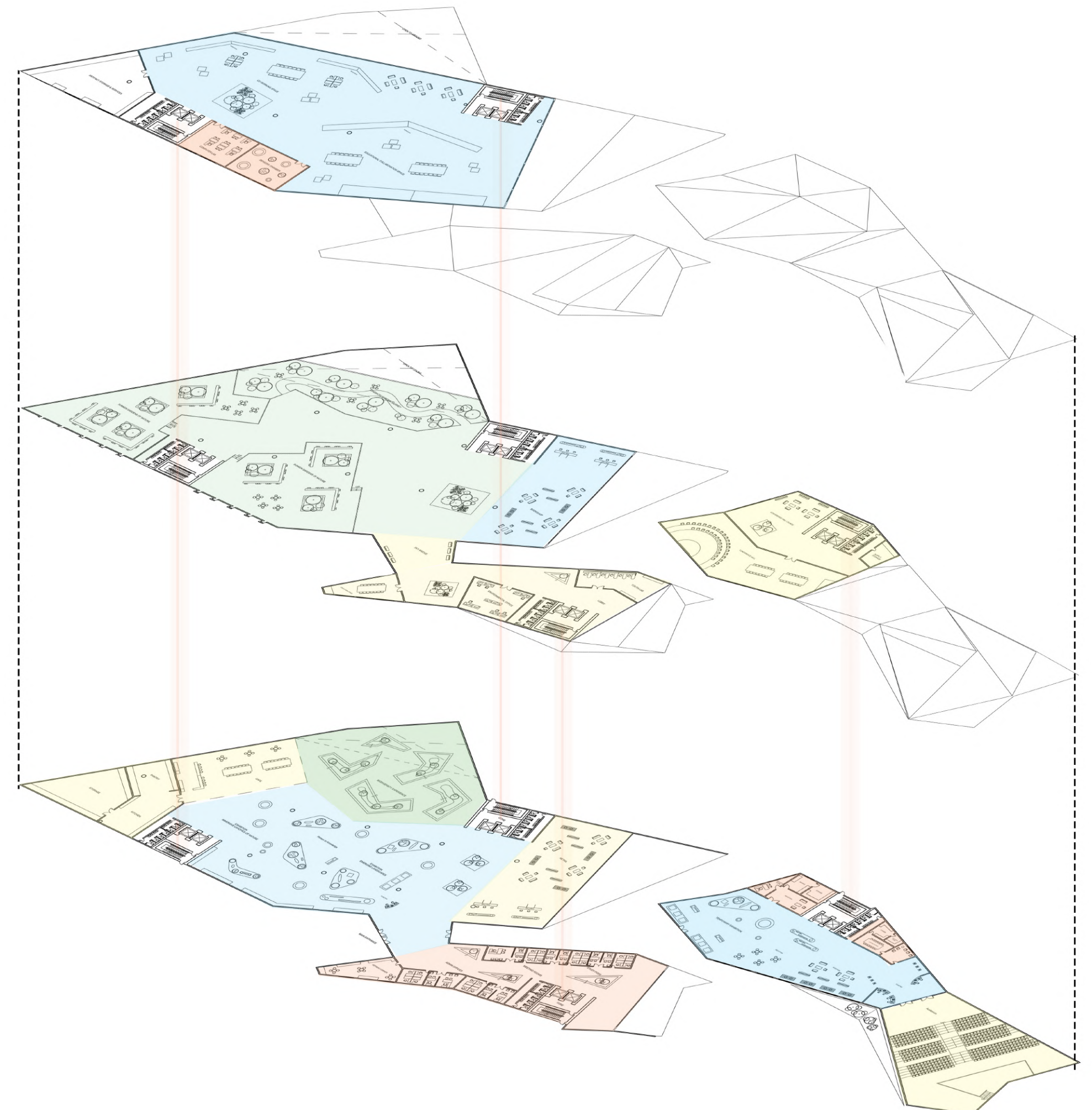




Order & Disorder of spaces and circulation from within

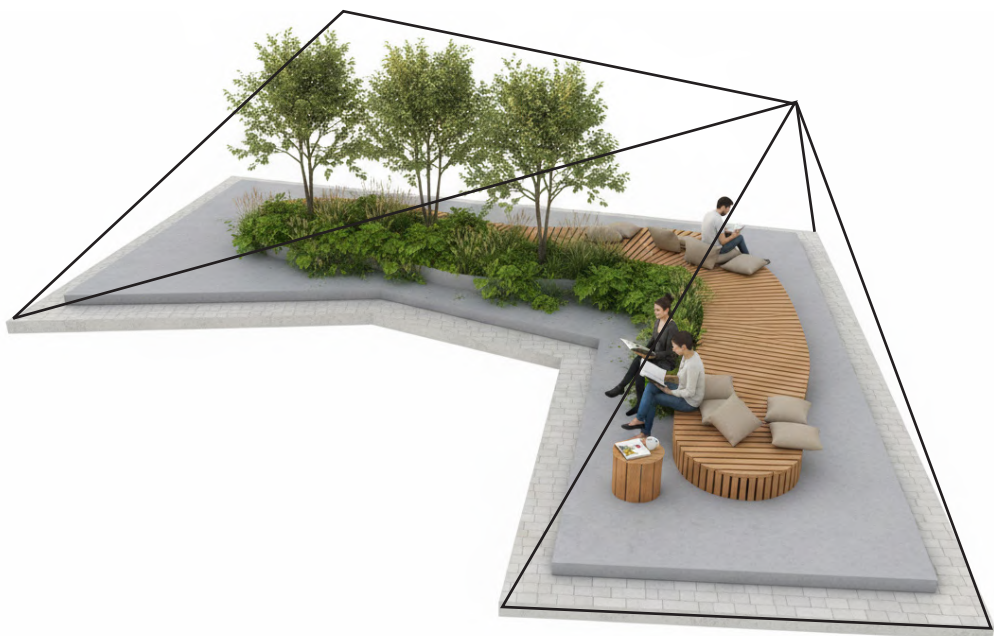
- Order
- Disorder
- Vertical Cores + Fire Escape

Order provides clarity, legibility, and structure through defined paths, organised program, and predictable movement, allowing users to understand how spaces connect and operate. Disorder, in contrast, allows movements of variation and freedom, where circulation expands or intersect encouraging exploration and different social and sensory experience. Together they create a dynamic architectural condition in which functional efficiency is balanced with spatial richness making circulation within the buiding understandable and engaging.

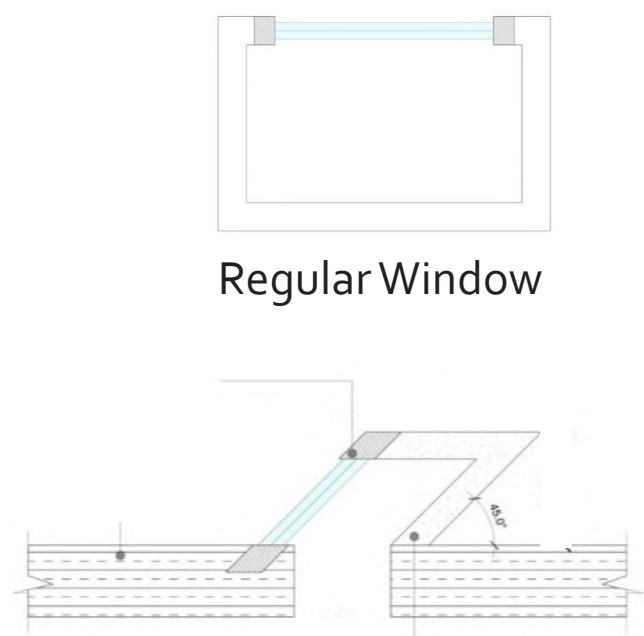


- Person to Self
- Person to Nature
- Person to Knowledge
- Person to Person

Person to Self, Person to Nature, Person to Knowledge, and Person to Person are relationships that are embedded within the architectural spatial qualities. Influencing how people move, gather, pause, and experience the space. Person to person is expressed through shared reading areas, informal seating zones, and gathering spaces that encourage interaction and exchange. Person to self is created through quieter corners and reflective spaces that allow privacy, focus, and personal retreat. Person to nature is integrated through light, planting, views, and natural materiality, strengthening the relationship between the interior and the surrounding environment. Person to knowledge is represented through the library's core function for intellect.



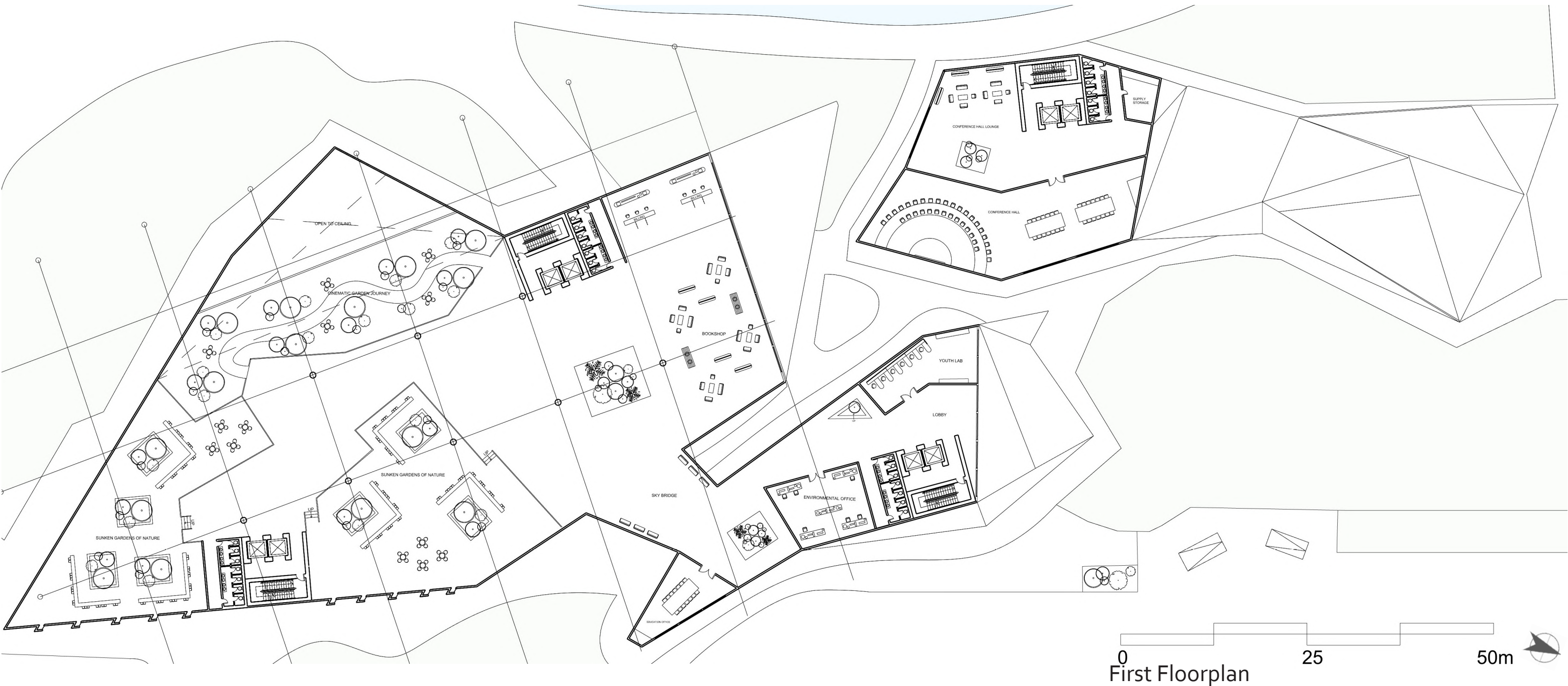
Biodiversity Pods Experience
Ground floor spatial qualities
element



Regular Window

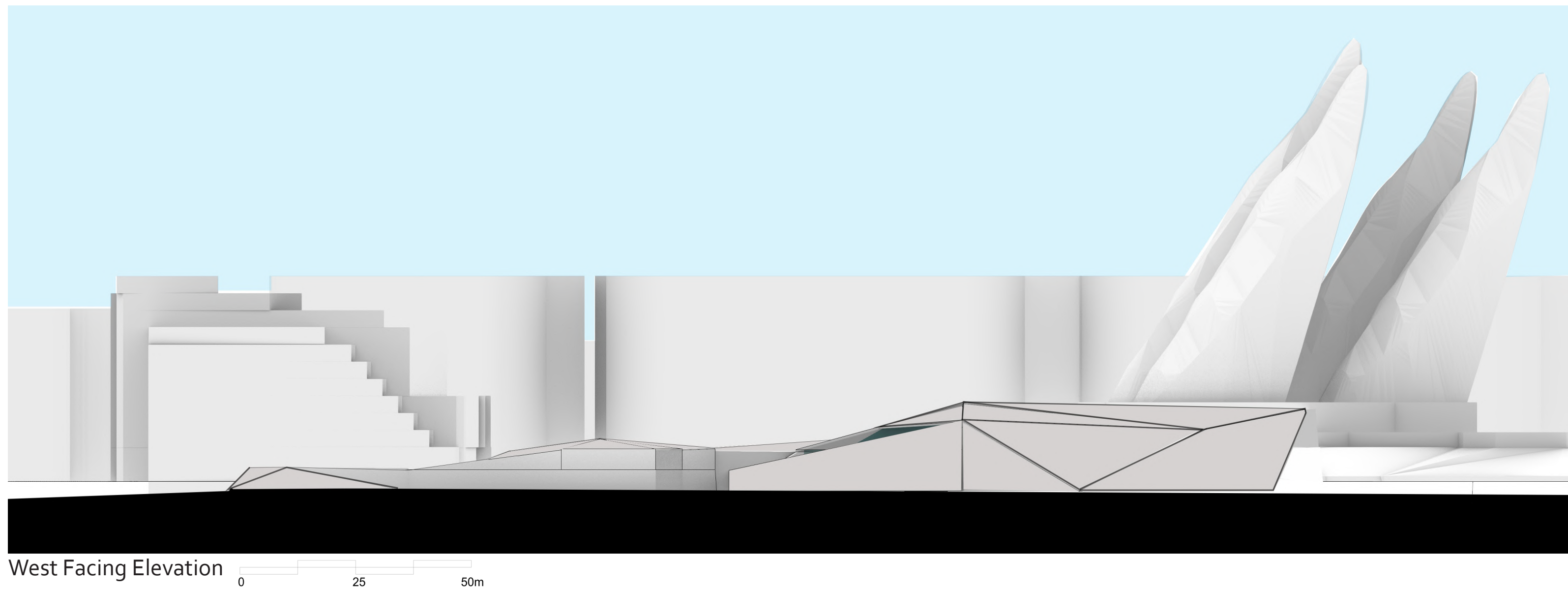
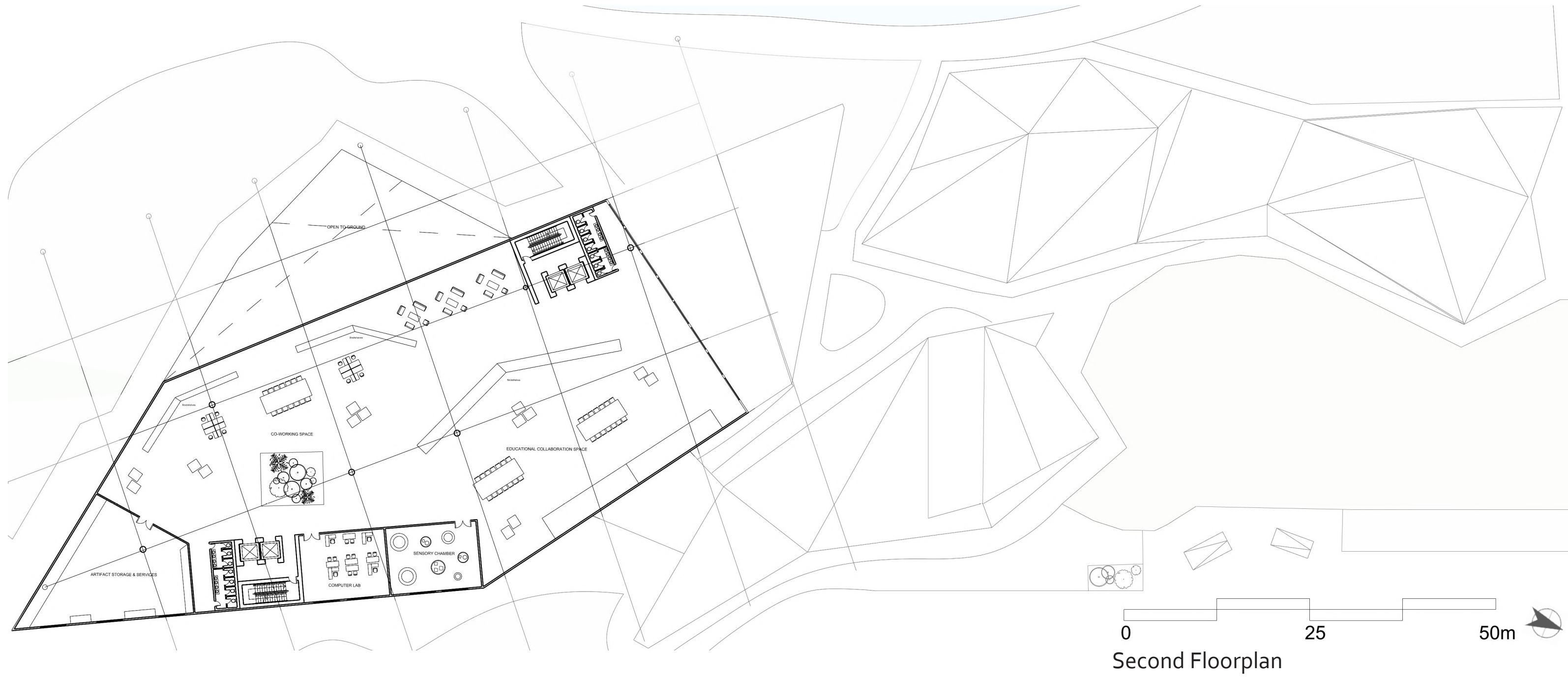
45 Degree Window Detail

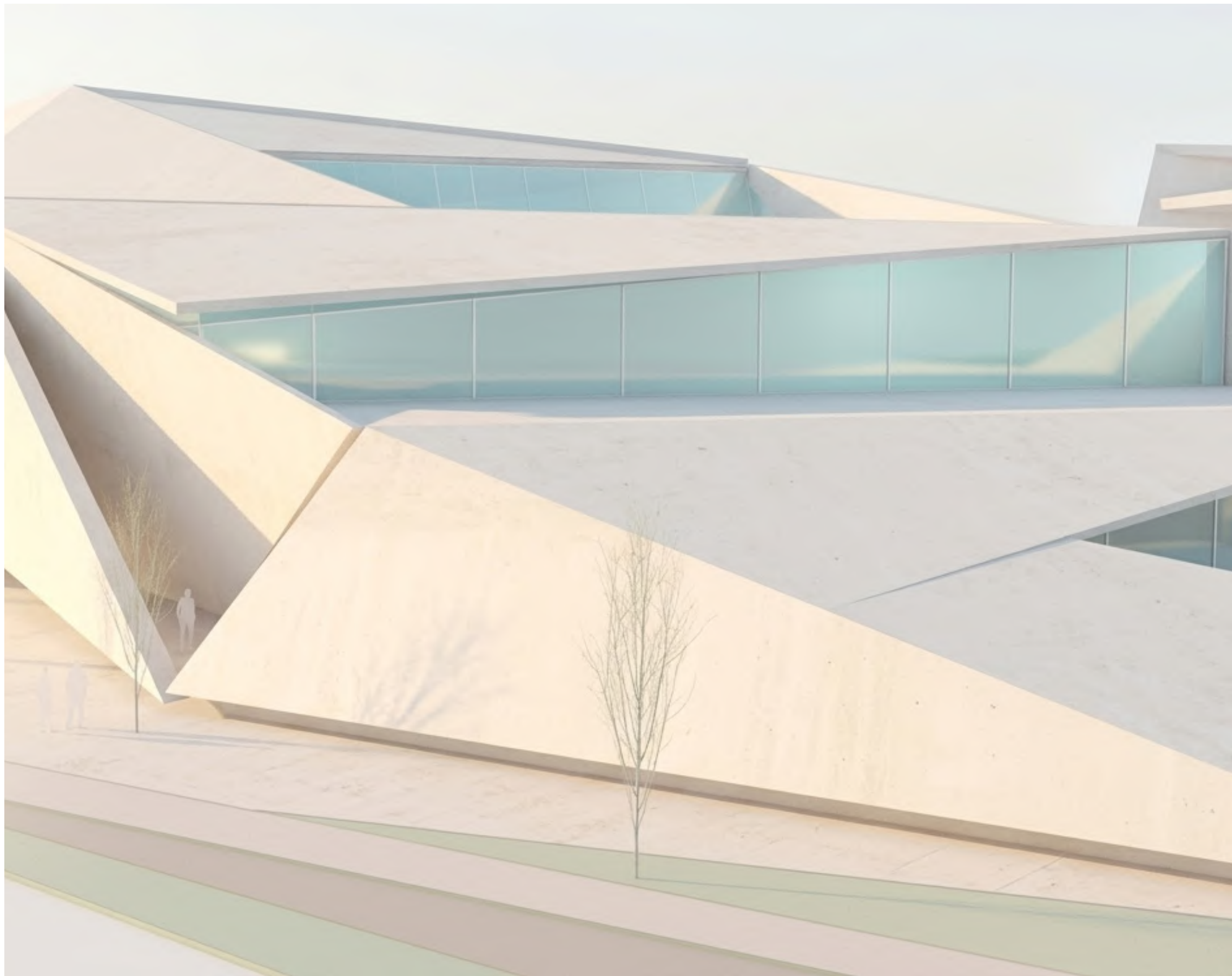
Rotated Windows can be installed to face North or East to minimize internal heat gains since the west and south facing recieve high levels of sunlight in the interior spaces.



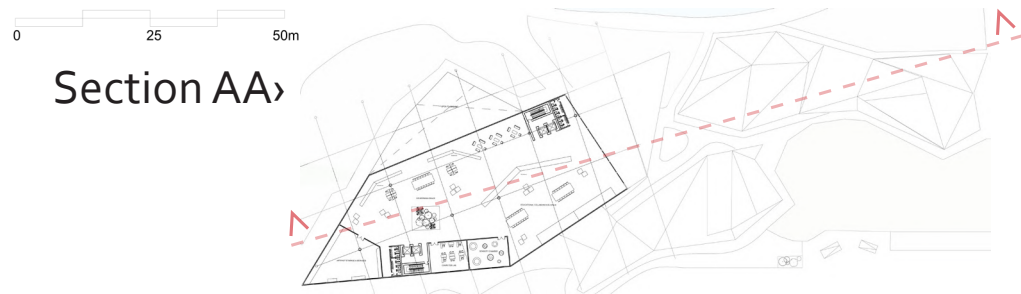
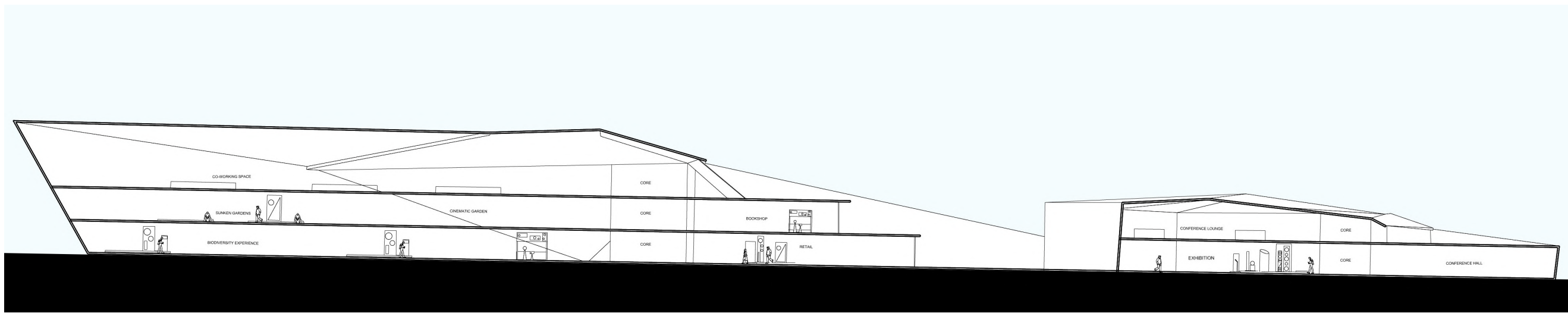


Person to Knowledge - Third Floor Educational Collaboration Space

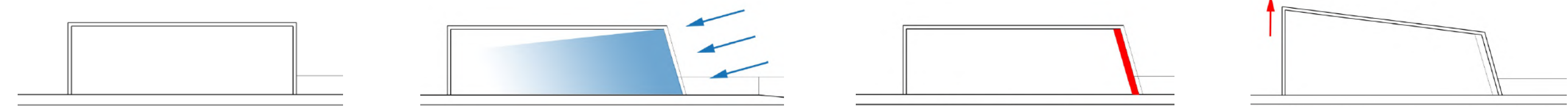




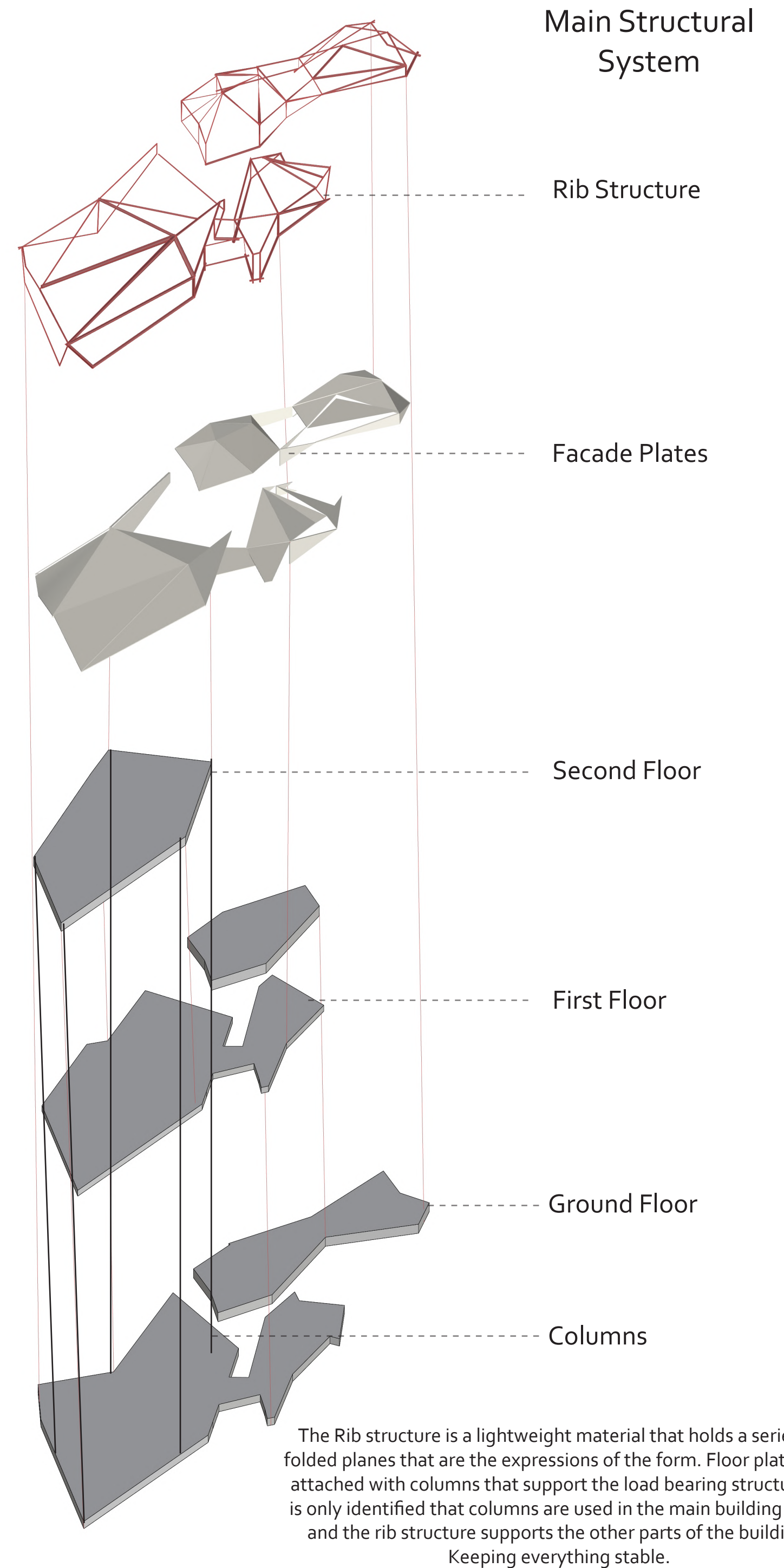
- Final Materials**
- Painted concrete
 - Glass
 - Steel

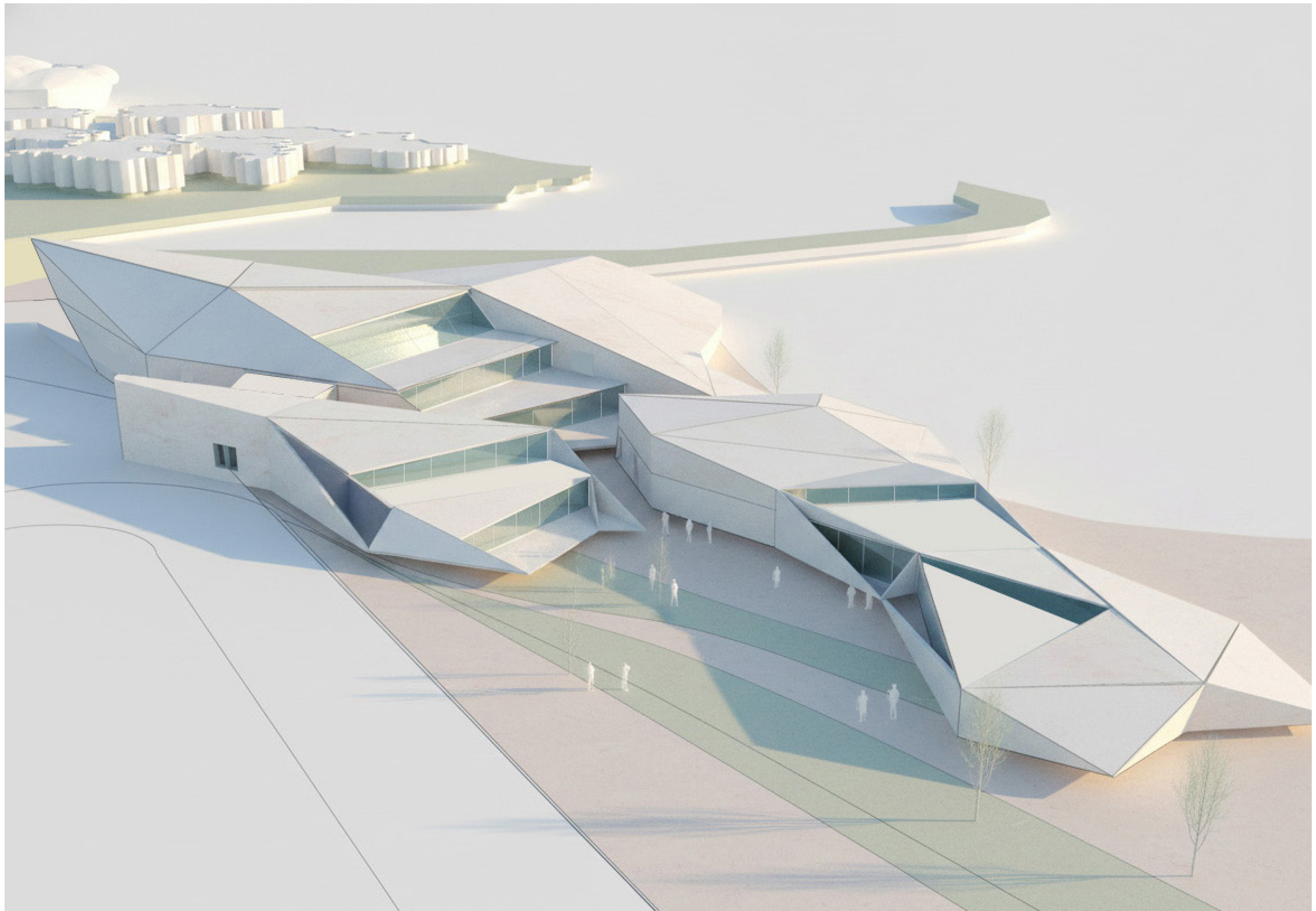


Sustainable elements within building design

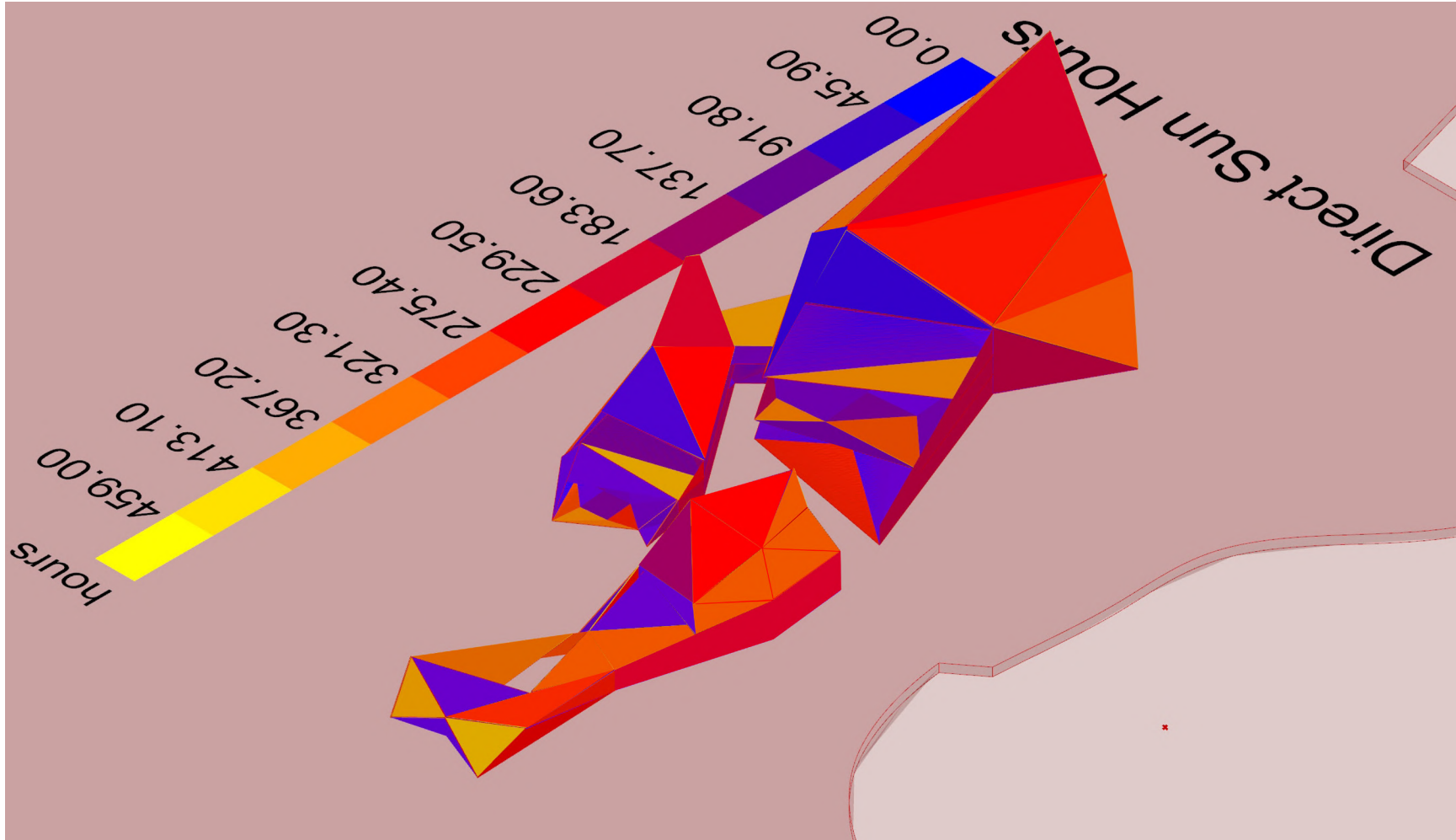


Starting in section the use of regular walls then slanting parts of the facade to capture the northern facing light, following the structure that is supported on the facade, finally raising and tapering the southern and western forms to respond to the climate.

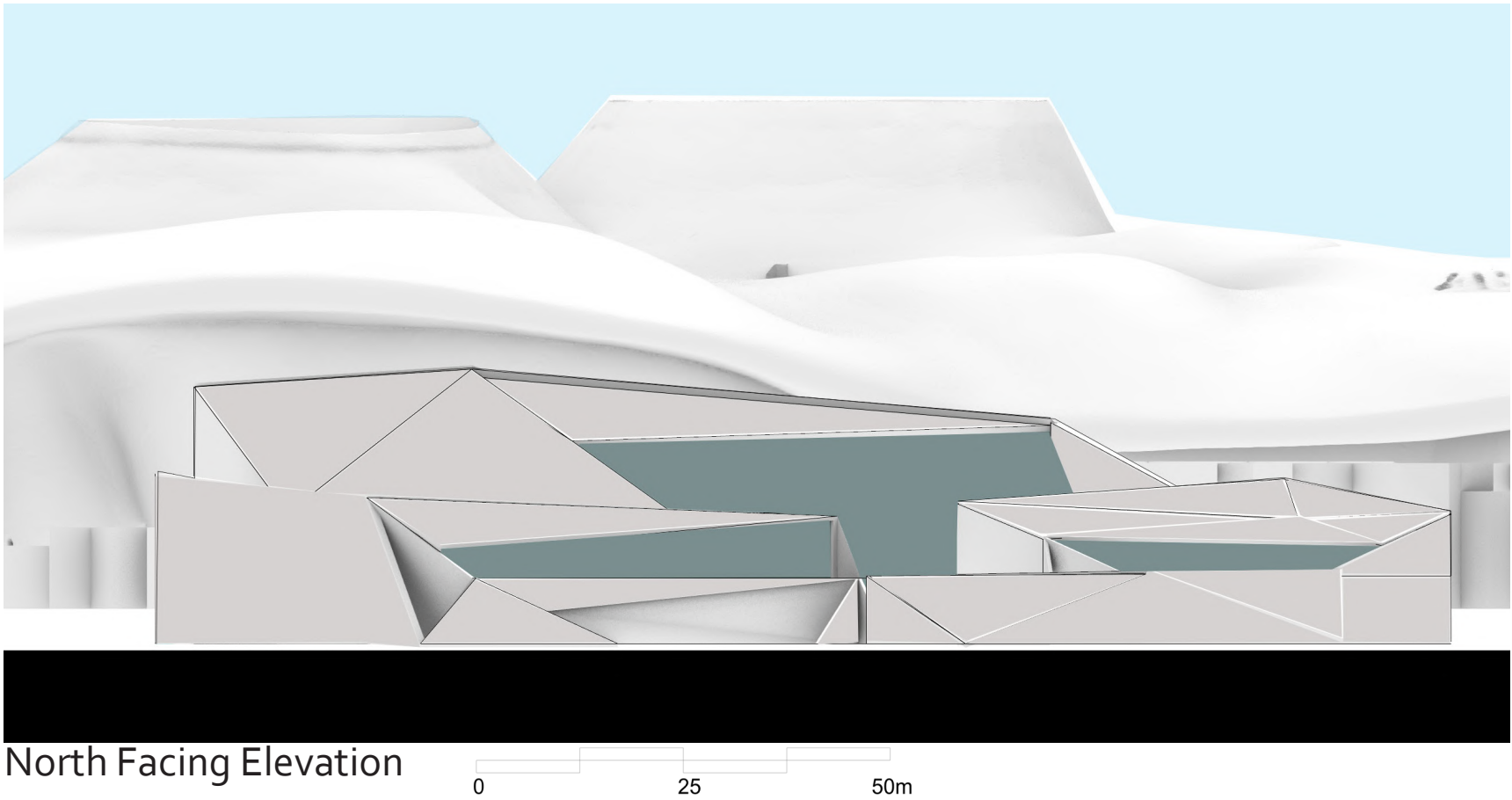




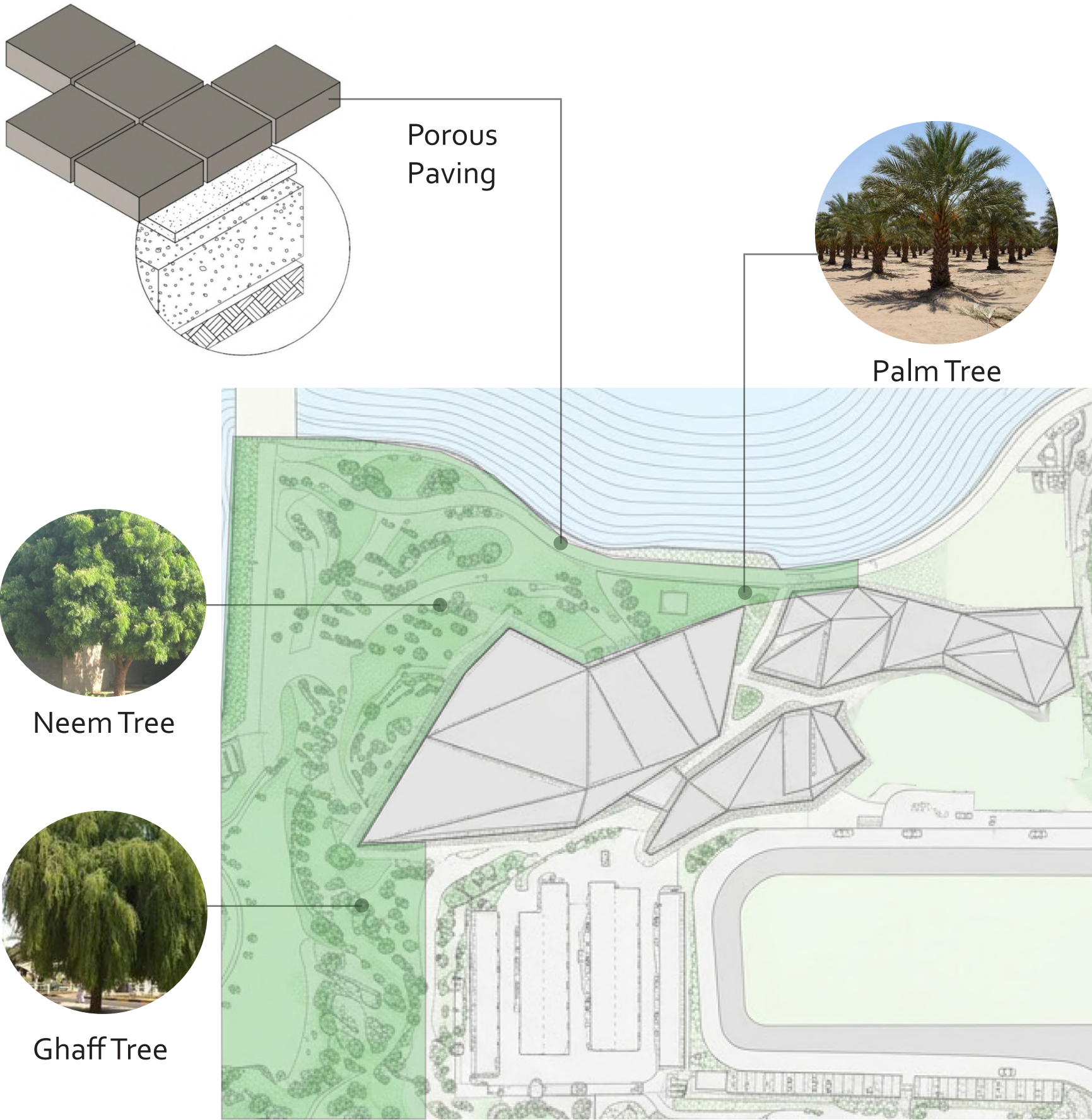
Environmental Elements & Sustainability



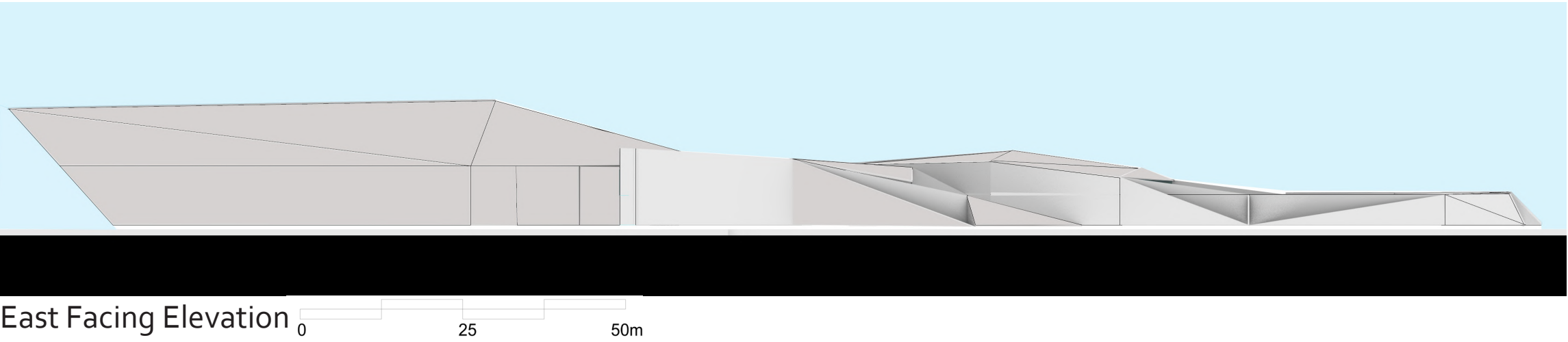
The intention of folding in architecture has lead my experimentation to the orientation of carving and solar performance. Which revealed that north facing windows recieve the least harsh direct sunlight and therefore experience lower heat gain. As a result the design responds by positioning the majority of openings toward the north, allowing for softer natural light while reducing solar exposure. This supports a more comfortable interior environment and strengthens the form .



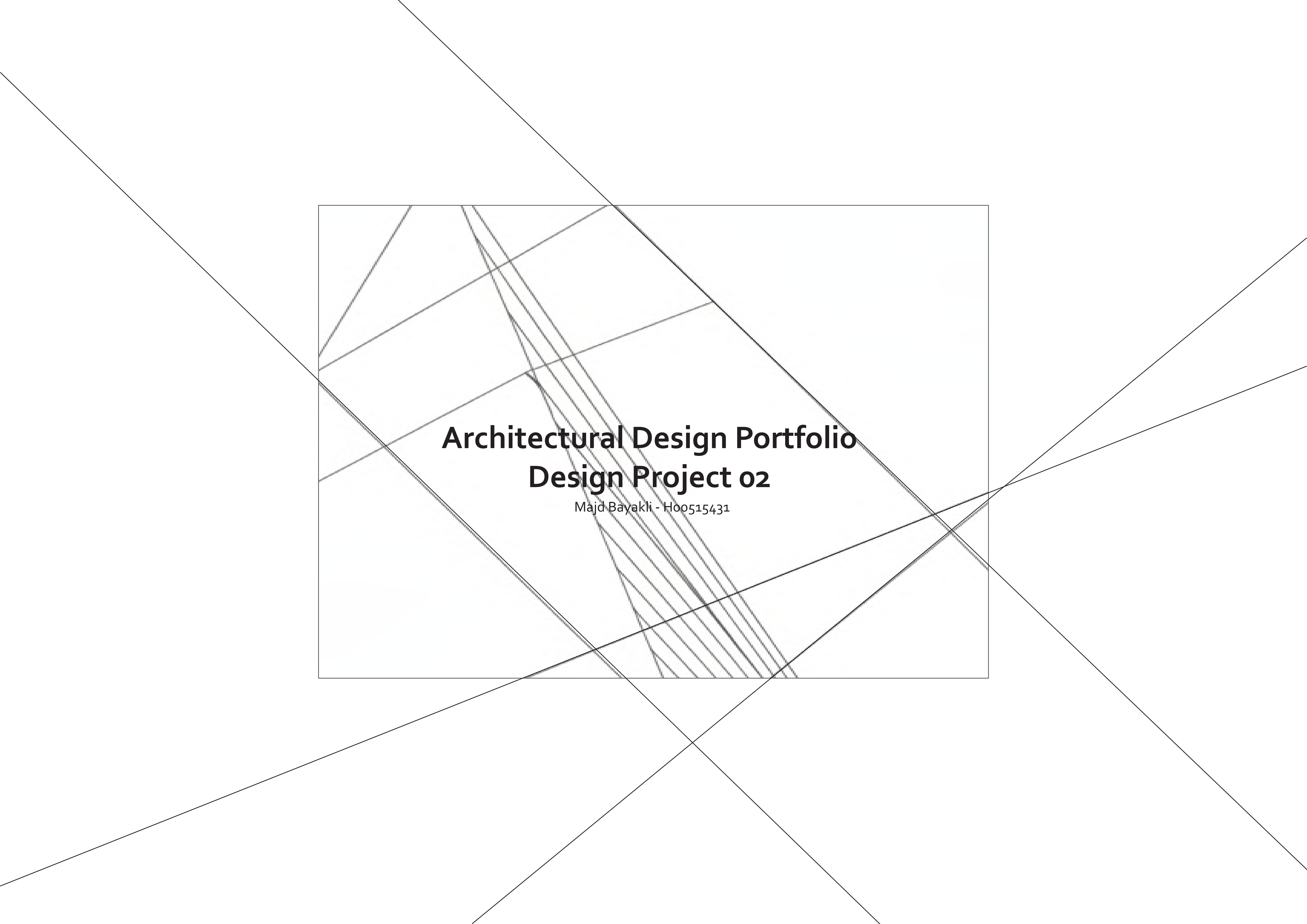
North Facing Elevation



Using local vegetation is a key role in the context of the deisgn, because it is naturally adapted to the hot, arid climate and requires less water, maintenance , and support than non native species. Using native plants are more resilient over time thus creating landscapes that respond more appropriately to the environmental conditions of the region.



East Facing Elevation



Architectural Design Portfolio

Design Project 02

Majd Bayakli - H00515431